



Federal Office
for Information Security

BSI Technical Guideline TR-03129-2

Protocols for the Management of Certificates and CRLs in Public-Key- Infrastructures (PKIs)

Part 2: Supplemental specifications for public and official authorities

Version 1.4.2



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1 Changelog

The following tables present the changes introduced between the latest versions of this technical guideline. The changelog lists the changes grouped per part of this Technical Guideline, per element (section, table, figure) and type of change, refer to [KeepAChangelog]:

- *Added* for new features
- *Changed* for changes in existing functionality
- *Deprecated* for soon-to-be removed features
- *Removed* for now removed features
- *Fixed* for any bug fixes
- *Security* in case of vulnerabilities

1.1 Changelog 1.4.2

TR Building Block	Type of Change	Change Description
Chapter Digital Seal Formats	Added	Added definitions for IDB as given in [ICAO TR IDB].
Signed Data Profiles for Signer Lists	Added	Added definitions for CSCA signer lists: • id-euHealthSealCscsList • id-icaoHealthSealCscsList • id-sovereignSealCscsList Added definitions for corresponding signer list signer certificates. Added criteria for signer list signer certificates. Added sections describing the creation and processing of signer lists.
Protocols for Digital Seals	Added	Added references for the CSCA signer lists: • id-euHealthSealCscsList • id-icaoHealthSealCscsList • id-sovereignSealCscsList
Chapter General Functions	Added	Added chapter for general functions with: • GetSupportedVersions • GetListStatus
Chapter Defect Lists for ePassport Application	Changed	Rename and adjust for signer certificates: • Document Signer Certificate Revoked to Signer Certificate Revoked • Document Signer Certificate incorrectly encoded or malformed to Signer Certificate incorrectly encoded or malformed
Chapter Defect Lists for ePassport Application	Changed	Adjusted for signer certificates: • Validity Period Incorrect • Wrong Signer Identifier • Issuing Country Defect
Chapter Defect Lists for ePassport Application	Changed	Rewrote text to include all signer certificates
Chapter Defect Lists for ePassport Application	Added	Added defect Certificate incorrectly encoded or malformed
Chapter Defect Lists for ePassport Application	Changed	Deprecated defect Document Signer Certificate incorrectly encoded or malformed

TR Building Block	Type of Change	Change Description
Chapter Defect Lists for ePassport Application	Changed	Redesigned reference of defect to certificate
Chapter Defect Lists for ePassport Application	Added	Added defect Distinguished Name Defect
Chapter Defect Lists for ePassport Application	Changed	Deprecated defect Issuing Country Defect
Chapter Signed Data Profiles for Signer Lists	Added	Check that signing time matches private key usage period of signer list signer certificate
Chapter Signed Data Profiles for Signer Lists	Changed	Reorganize requirements regarding Signer List Signer, creating and processing Signer Lists.

Table 1.1 Changelog

1.2 Changelog 1.4.1

TR Building Block	Type of Change	Change Description
Chapter Defect Lists for ePassport Application	Added	Added definition of Defect List Signer OID for extendedKeyUsage in Defect List Signer certificates
Chapter Defect Lists for ePassport Application	Added	Added mandatory check for systems using Defect Lists
Chapter Defect Lists for ePassport Application	Changed	Clarified that a defective certificate is only one example for the rationale of adding a hash to a defect
Chapter Signed Data Profiles for Signer Lists	Added	Added definitions for signers used for certificate exchange
Chapter Signed Data Profiles for Signer Lists	Added	Added regulation for the use of those signers in the corresponding lists.
Chapter Signed Data Profiles for Signer Lists	Added	Added definition of DocumentSigner List Signer OID for extendedKeyUsage in DocumentSigner List Signer certificates.

Table 1.2 Changelog

1.3 Changelog 1.4.0

TR Building Block	Type of Change	Change Description
Whole document	Changed	Updated references: • TR-03129 to TR-03129-1 and TR-03129-3 • TR-03110 to TR-03110-1 and TR-03110-3 • ICAO Doc 9303 Seventh Edition to ICAO Doc 9303 Eighth Edition
Accompanying WSDL files	Changed	Changed structure to achieve "once only" for type definitions
Accompanying WSDL files	Added	Added DigitalSeals folder and files corresponding to the new chapter Protocols for Digital Seals
Chapter Introduction	Changed	Revised the chapter.
Chapter Implementation as Web Services	Removed	Removed chapter because it was integrated into the other chapters.
Chapter Protocols for Passive Authentication	Changed	Revised chapter: • Changed descriptions. • Changed the names of some parameters to allow references from the chapter Protocols for Digital Seals and to streamline the terminology.

TR Building Block	Type of Change	Change Description
Chapter Protocols for Passive Authentication	Changed	Added new parameters per defect returned by the TCC for GetDocumentSignerInformation: • "applied" allows a TCC to indicate if it already applied a Defect. • "preDefectResult" allows a TCC to communicate the original result of a check, before the corresponding defect was applied.
Chapter Protocols for Terminal Authentication	Added	Added specification for Terminal Authentication in distributed Terminals originating from TR-03129, i.e. • GetCertificateChain • GetTASignature
Chapter Protocols for Terminal Authentication	Added	Added message RequestForeignCertificate originating from TR-03129
Chapter Protocols for Terminal Authentication	Added	Added message GetDVCertificates
Chapter Protocols for Digital Seals	Added	Added chapter to support checking digital seals Added the following messages: • GetSignerList • SendSignerList • GetSignerCertificate • GetDigitalSealVerification
Chapter Digital Seal Formats	Added	Added chapter to support checking digital seals. Added descriptions for the following formats: • VDS • VDSNC • EUCWT
Chapter Defect Lists for ePassport Application	Changed	Revised introduction.
Chapter Defect Lists for ePassport Application	Changed	Updated RFC3852 reference to RFC5652 reference.
Chapter Defect Lists for ePassport Application	Changed	Changed details of the following defects: • Document Signer Certificate Malformed : Added ASN.1 Definition for parameter ReplacementCertificate • Authentication Protocol Failure : Added more status codes • Wrong Signer Identifier : Added ASN.1 Definition for parameter correctedSignerIdentifier • Card Security Object Malformed : Added ASN.1 Definition for parameter correctedCardSecurityObject
Chapter Signed Data Profile for Signer Lists	Added	Added specifications for new signer lists: • id-euHealthSealSignerList • id-icaoHealthSealSignerList • id-sovereignSealSignerList
Chapter WSDL and XML Scheme specifications	Removed	Removed chapter because it became unnecessary.
Appendix A	Added	Added content from former TR-03129.

Table 1.3 Changelog

2 Introduction

This technical guideline specifies Public Key Infrastructure (PKI)-related communication protocols for security mechanisms in the context of Machine Readable Travel Documents (MRTDs). This guideline is part of the BSI TR-03129 series and depends on the basic document [BSI-TR-03129-1] and on the PKI-related communication protocols for security mechanisms of Electronic Identity Documents (eIDs) based on Extended Access Control (EAC) described in [BSI-TR-03129-3].

The target audience for this document are public services and official authorities that want to implement an Inspection System (IS) for MRTDs and travel related documents. This guideline describes the PKI-related communication inside a distributed terminal (see ▶Appendix A) and between the members of a background service infrastructure. Implementing the protocols described in this Technical Guideline (TR) enables verifying Electronic Machine Readable Travel Documents (eMRTDs) and travel related documents with digital seals.

2.1 Terminology

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in [RFC2119] when, and only when, they appear in capital letters.

The key word "CONDITIONAL" is to be interpreted as follows: The usage of an item depends on the use of other items. It is therefore further qualified under which conditions the item is REQUIRED or RECOMMENDED.

2.2 Roles

2.2.1 Reader

A reader contains a Proximity Coupling Device (PCD) used for communication with the Integrated Circuit (IC) of an eMRTD.

2.2.2 Hardware Security Module (HSM)

A Hardware Security Module (HSM) offers secure storage for digital keys.

2.2.3 Terminal

A terminal consists of at least one reader and one secure key store, usually a HSM.

2.2.3.1 Distributed Terminal

A terminal with a Terminal Control Centre (TCC). All readers belonging to the distributed terminal need to have a permanently active connection to the TCC.

2.2.4 Terminal Control Centre (TCC)

A part of a distributed terminal. It manages a secure key storage (usually a HSM) and offers a communication interface to all readers that belong to the distributed terminal. It MUST be assured that a TCC only communicates with known readers.

2.2.5 Document Verifier (DV)

An entity in the background infrastructure. Above the TCC in the hierarchy. Below the Country Verifying Certificate Authority (CVCA) respectively National Public Key Directory (NPKD) in the hierarchy.

2.2.6 Country Verifying Certificate Authority (CVCA)

An entity in the background infrastructure. Above the Document Verifier (DV) in the hierarchy.

2.2.7 National Public Key Directory (NPKD)

An entity in the background infrastructure. Above the DV in the hierarchy. The NPKD is the national key store for known certificates in the context of MRTDs.

2.3 Procedures

Data stored on eMRTDs can be proven authentic with Passive Authentication (PA) (see [ICAO Doc 9303 p11]). To execute PA an IS needs access to Country Signing Certificate Authority (CSCA) certificates and Document Signer (DS) certificates.

eMRTDs can be protected against unauthorized access with Terminal Authentication (TA) (see [BSI-TR-03110-1]). To execute TA an IS needs access to CVCA certificates, DV certificates, terminal certificates and a matching private terminal key.

In a distributed terminal, access to certificates and private keys is provided to all connected readers via the TCC. Hence, any TA triggers a request to the TCC which then passes cryptographic operations to the secure key store. Regardless at which reader of a distributed terminal an eMRTD is presented, an identical key pair and Card Verifiable (CV) certificates are used for TA between the distributed terminal and an eMRTD.

2.4 Communication Model

To communicate between sender and receiver, [BSI-TR-03129-1] defines and describes:

- Messages
- Common Parameters
- Common Return Codes

The authentication of the sender and receiver of messages as well as the confidentiality and integrity of the contents of the messages are not considered at the level of these messages. Recommendations are described in [BSI-TR-03129-1], see “Authentication, Confidentiality, and Integrity Protection”.

2.5 Publication Language

This technical guideline is only published in English. The reasons for this are as follows:

- Relation to other English only publications (esp. ICAO Doc 9303, RFC5652, EU DCC, TR-03135, TR-03137)
- The content is relevant in an European context because of the CVCA PKI infrastructure

3 General Functions

This chapter defines general functions that are independent of the supported protocols.

3.1 Implementation as Web Services

It is RECOMMENDED that the messages defined in this chapter are realized using the Simple Object Access Protocol (SOAP) messages described in the attached WSDL specifications.

3.2 Acquiring general information

The following message is used by readers to check which versions of this technical guideline are supported by the TCC.

GetSupportedVersions ▶Section 3.2.1

3.2.1 GetSupportedVersions

This message is used to get a list of supported versions of this TR. Its parameters are given in ▶Table 3.1.

Parameter	Description
Input	
None	This message has no input.
Output	
supportedVersions	REQUIRED A list of the versions of this TR.
returnCode	REQUIRED The following return codes are possible: • ok_list_available • failure_internal_error • failure_other_error • failure_syntax
returnCodeMessage	OPTIONAL Further information regarding the processing of this message.

Table 3.1 Input and output for the message GetSupportedVersions

3.2.2 GetListStatus

This message is used to get status information about the lists used by the TCC. Its parameters are given in ▶Table 3.2. It is RECOMMENDED to use this message before performing document inspections according to [BSI-TR-03135-1].

Parameter	Description
Input	
None	This message has no input.
Output	
listStatusInformation	CONDITIONAL A list of listStatus elements. The format of a listStatus is described in ▶Table 3.3. REQUIRED if there are lists present in the TCC, SHALL be omitted otherwise.

Parameter	Description
returnCode	REQUIRED The following return codes are possible: • ok_available • ok_no_lists • failure_internal_error • failure_other_error • failure_syntax
returnCodeMessage	OPTIONAL Further information regarding the processing of this message.

Table 3.2 Input and output for the message GetListStatus

Parameter	Description
type	REQUIRED The Object Identifier (OID) of the list.
hash	REQUIRED The SHA-256 hash of the list.

Table 3.3 ListStatus (ListStatusType) definition

4 Protocols for Passive Authentication

All protocols and specifications for PA SHALL be implemented as specified in [BSI-TR-03129-3] with the following exceptions:

- References to the role "Terminal" SHALL be interpreted as references to the role "TCC".
- References to the role "CVCA" SHALL be interpreted as references to the role "NPKD".
- The role "TCC" SHALL implement an additional Web Service towards its connected readers supporting the following messages:
 - GetMasterList (see [BSI-TR-03129-3])
 - GetDefectList (see [BSI-TR-03129-3])
 - GetDocumentSignerList (▶Section 4.2.1)
 - GetDocumentSignerInformation (▶Section 4.3.1)

In contrast to the specifications given in [BSI-TR-03129-3], this Web Service SHALL only support synchronous communication: Hence, the input parameter `callbackIndicator` MUST be set to `callback_not_possible`. Furthermore parameters `messageID` and `responseURL` MUST be omitted.

- The role "TCC" SHALL implement the additional message `SendDocumentSignerList` (▶Section 4.2.2) for the Web Service towards the role "DV".
- The role "DV" SHALL implement the additional message `GetDocumentSignerList` (▶Section 4.2.1) for the Web Service towards the role "TCC".
- The role "DV" SHALL implement the additional message `SendDocumentSignerList` (▶Section 4.2.2) for the Web Service towards the role "NPKD".
- The role "NPKD" SHALL implement the additional message `GetDocumentSignerList` (▶Section 4.2.1) for the Web Service towards the role "DV".

4.1 Implementation as Web Services

It is RECOMMENDED that the messages defined in this chapter are realized using the SOAP messages described in the attached WSDL specifications.

4.2 Distribution of Document Signer Lists / DS Certificates

The following messages are used for the distribution of Document Signer Lists / DS certificates:

- GetDocumentSignerList ▶Section 4.2.1
- SendDocumentSignerList ▶Section 4.2.2

These messages enable the distribution of signer certificates to facilitate executing PA for eMRTDs without a signer certificate inside the Elementary File Document Security Object (EF.SOD). This is necessary because in an older version of the ICAO Logical Data Structure (LDS) for eMRTDs (see [ICAO Doc 9303 p10]) the presence of a signer certificate inside the EF.SOD was not required.

4.2.1 GetDocumentSignerList

This message is used to get a signed list of DS certificates. Its parameters are given in ▶Table 4.1.

Parameter	Description
Input	
None	This message has no input.
Output	

Parameter	Description
documentSignerList	CONDITIONAL A signed list of DS certificates. It is REQUIRED if the message was processed without errors. It SHALL be missing otherwise.
returnCode	REQUIRED The following return codes are possible: • ok_list_available • failure_list_not_available • failure_internal_error • failure_other_error • failure_syntax
returnCodeMessage	OPTIONAL Further information regarding the processing of this message.

Table 4.1 Input and output for the message GetDocumentSignerList

4.2.2 SendDocumentSignerList

This message is used to send a signed list of DS certificates. Its parameters are given in ►Table 4.2.

Parameter	Description
Input	
documentSignerList	REQUIRED A signed list of DS certificates.
Output	
returnCode	REQUIRED The following return codes are possible: • ok_received_correctly • failure_internal_error • failure_other_error • failure_syntax
returnCodeMessage	OPTIONAL Further information regarding the processing of this message.

Table 4.2 Input and output for the message SendDocumentSignerList

4.3 Passive Authentication for distributed terminals

For PA, a reader must validate the DS certificate and the signature of one or more Security Objects. For this, the DS certificate has to be validated against the corresponding, trusted CSCA certificate distributed via a trusted Master List (ML).

The message GetDocumentSignerInformation ►Section 4.3.1 is used as part of PA within a distributed terminal.

The readers that are connected to a TCC inside a distributed terminal use this message in the process of executing PA. Therefore, the processing of this message is time-critical and MUST be done synchronously.

4.3.1 GetDocumentSignerInformation

As the CSCA certificates are stored securely in the TCC, a connected reader sends this message to the TCC to get all information associated with the DS certificate. This information includes the validity of the DS certificate as well as other information used for PA. The parameters of the message GetDocumentSignerInformation are given in ►Table 4.3.

Parameter	Description
Input	
signerCertificate	CONDITIONAL It is REQUIRED if a signer certificate is included inside the Security Object. The signer certificate SHALL be provided as it is encoded in the Security Object.
signerIdentifier	CONDITIONAL It is REQUIRED if the parameter signerCertificate is not present, otherwise RECOMMENDED. The signerIdentifier SHALL be provided as it is encoded in the Security Object.
returnCACertificate	REQUIRED Indicates whether the caller wants to receive the CA certificate corresponding to the signer certificate contained in signerCertificate or indicated by signerIdentifier.
Output	
signerInformation	CONDITIONAL It is REQUIRED if the message is processed successfully. It SHALL be omitted otherwise. The structure of the parameter is given in ▶Table 4.4.
returnCode	REQUIRED The following return codes are possible: • ok_syntax • failure_internal_error • failure_other_error • failure_syntax
returnCodeMessage	OPTIONAL Further information regarding the processing of this message.

Table 4.3 Input and output for the message GetDocumentSignerInformation

The parameter signerInformation contains more information about the signer certificate. The parameter has the type signerInformationType and its structure is given in ▶Table 4.4.

Parameter	Description
trustStatus	REQUIRED The TCC's rating for the given signer certificate. See [BSI-TR-03135-1] section 4.6.4.3.6 for details regarding trust status. The following values are possible: • certificate_ok • certificate_undetermined • certificate_invalid
certificateChainValidity	REQUIRED Information about the validity of the certificate chain. The structure of the parameter is given in ▶Table 4.5.
signerCertificateSignature	REQUIRED Result of the validation of the signature of the signer certificate. The structure of the parameter is given in ▶Table 4.6.

Parameter	Description
knownDefects	CONDITIONAL A list of defects for the given signer certificate known to the TCC. The parameter is REQUIRED if the TCC knows defects for the given signer certificate. It SHALL be omitted otherwise. The structure of a single entry knownDefect is given in ▶Table 4.8.
caCertificate	CONDITIONAL In case of returnCACertificate being true, this parameter contains the base64-encoded Certificate Authority (CA) certificate which has issued the signer certificate, as given in the Master List.
signerCertificate	CONDITIONAL In case of signerIdentifier being set, this parameter contains the base64-encoded signer certificate which has been used to sign the Security Object.
unknownCriticalExtension	CONDITIONAL Unknown critical extensions contained in the DS certificate. It is REQUIRED if such extensions exist. It SHALL be omitted otherwise.
listStatusInformation	CONDITIONAL A list of listStatus elements. The format of a listStatus is described in ▶Table 3.3. REQUIRED if there are lists present in the TCC, SHALL be omitted otherwise.

Table 4.4 signerInformation (signerInformationType) definition

The parameter certificateChainValidity contains validity information about the signer certificate and the corresponding CA certificate. The parameter has the type certificateChainValidityType and its structure is given in ▶Table 4.5.

Parameter	Description
caCertificateValidity	REQUIRED Validity information of the CA certificate. The structure of the parameter is given in ▶Table 4.7.
signerCertificateValidity	REQUIRED Validity information of the signer certificate. The structure of the parameter is given in ▶Table 4.7.

Table 4.5 certificateChainValidity (certificateChainValidityType) definition

The parameter signerCertificateSignature contains trust status information about the signer certificate. The parameter has the type signatureType, whose structure is given in ▶Table 4.6.

Parameter	Description
status	REQUIRED See [BSI-TR-03135-1] for details regarding signature checking. The following values are possible: • signature_ok • signature_invalid • signature_not_checked
message	OPTIONAL Further information regarding the signature.

Table 4.6 signerCertificateSignature (signatureType) definition

Information about the validity of a certificate (e.g. caCertificateValidity) is given by the type certificateValidityType, whose structure is given in ▶Table 4.7.

Parameter	Description
status	REQUIRED The evaluation result of the validity period according to the time of the TCC. See [BSI-TR-03135-1] for details regarding validity period evaluation (for the example CSCA and DS certificate). The following values are possible: • certificate_valid • certificate_expired • certificate_not_available • certificate_not_yet_valid
effectiveDate	CONDITIONAL Effective date of the certificate. It SHALL be omitted if status is certificate_not_available. It is REQUIRED otherwise.
expirationDate	CONDITIONAL Expiration date of the certificate. It SHALL be omitted if status is certificate_not_available. It is REQUIRED otherwise.
message	OPTIONAL Further information regarding the certificate validity.

Table 4.7 certificateValidityType definition

The parameter knownDefect contains information about the application of defect. The parameter has the type knownDefectType, whose structure is given in ▶Table 4.8.

Parameter	Description
critical	REQUIRED Indicates whether defect is critical.
applied	REQUIRED Indicates whether defect was already applied by the TCC.
preDefectResult	CONDITIONAL Result before defect was applied by the TCC. It is REQUIRED if applied is true and the condition for at least one of the subparameters of this parameter is fulfilled. It SHALL be omitted otherwise. The structure of the parameter is given in ▶Table 4.9. See [BSI-TR-03135-1] for details regarding the effects of defects.
defect	REQUIRED A base64-encoded VersionedKnownDefect (see ▶Section 8.1).

Table 4.8 knownDefect (knownDefectType) definition

Information about the results before defect was applied are given by preDefectResult. The parameter has the type preDefectResultType, whose structure is given in ▶Table 4.9.

Parameter	Description
trustStatus	CONDITIONAL Trust status before the defect was applied. It is REQUIRED if this value was changed by applying the defect. It SHALL be omitted otherwise. The possible values are given in ▶Table 4.4 parameter "trustStatus".

Parameter	Description
signatureStatus	CONDITIONAL Signature status status before the defect was applied. It is REQUIRED if this value was changed by applying the defect. It SHALL be omitted otherwise. The possible values are given ▶Table 4.6 parameter "status".
certChainValidity	CONDITIONAL Certificate chain validity status before the defect was applied. It is REQUIRED if this value was changed by applying the defect. It SHALL be omitted otherwise. The structure of the parameter is given in ▶Table 4.5.

Table 4.9 preDefectResult (preDefectResultType) definition

5 Protocols for Terminal Authentication

All protocols and specifications for Terminal Authentication SHALL be implemented as specified in [BSI-TR-03129-3] with the following exceptions:

- References to the role "Terminal" SHALL be interpreted as references to the role "TCC"
- The role "TCC" SHALL implement an additional Web Service towards its connected readers supporting the following messages:
 - GetCertificateChain (▶Section 5.2.1)
 - GetTASignature (▶Section 5.2.2)
- The role "CVCA" SHALL implement the following additional messages for the Web Service towards the role "DV":
 - GetDVCertificates (▶Section 5.3.1).
 - RequestForeignCertificate (▶Section 5.4.1)

5.1 Implementation as Web Services

It is RECOMMENDED that the messages defined in this chapter are realized using the SOAP messages described in the attached WSDL specifications.

5.2 Terminal Authentication for distributed terminals

The following messages are used as part of TA within a distributed terminal:

- GetCertificateChain (▶Section 5.2.1)
- GetTASignature (▶Section 5.2.2)

The readers that are connected to a TCC inside a distributed terminal use these messages in the process of executing TA. Therefore, the processing of these messages is time-critical and MUST be performed synchronously.

5.2.1 GetCertificateChain

To execute TA, a reader must obtain the CV certificate of the TCC and a complete chain of certificates up to a CVCA CV certificate that is known to the eMRTD the reader is communicating with. This message allows a reader to download all CV certificates from the TCC that are needed for executing TA with a specific eMRTD.

The parameters of the message GetCertificateChain are given in ▶Table 5.1.

Parameter	Description
Input	
keyCAR	REQUIRED A Certificate Authority Reference (CAR) obtained from an eMRTD. This is a reference to one of the public keys which can be used by the eMRTD for the verification of CV certificates.
Output	
certificateSeq	CONDITIONAL If the message is processed successfully, this parameter SHALL contain all valid link certificates of the CVCA identified by the CAR contained in the parameter keyCAR, the CV certificate of the DV derived from the most recent CVCA certificate and the corresponding CV certificate of the TCC.

Parameter	Description
returnCode	REQUIRED The following return codes are possible • ok_certificate_chain_available • failure_CAR_unknown • failure_internal_error • failure_other_error • failure_syntax
returnCodeMessage	OPTIONAL Further information regarding the processing of this message.

Table 5.1 Input and output for the message GetCertificateChain

5.2.2 GetTASignature

In a distributed terminal, the private key for TA is stored in a secure key store within the TCC. With this message a reader requests the calculation of a signature for TA. The calculation of this signature is defined in [BSI-TR-03110-1]. There it is denoted as s_{PCD} .

This message MAY be used in the following two ways:

- First option: The reader calculates a hash value ($hashTBS$) over the data to be signed and sends this hash value to the TCC.
- Second option: The reader sends all data to be signed to the TCC. The TCC calculates the hash value of the data before signing it.

The first option SHOULD be used, if an eMRTD can be uniquely identified with a part of the data to be signed (i.e. with the value of $idPICC$ and/or $challengePICC$). When using the first option, this possibly sensitive information is only held in the reader's volatile storage.

The parameters of GetTASignature are given in ►Table 5.2.

Parameter	Description
Input	
keyCHR	REQUIRED Identification of the private key to be used for the calculation of the signature. The Certificate Holder Reference (CHR) of the corresponding CV certificate SHALL be given in the parameter.
hashTBS	CONDITIONAL The hash value calculated over the data to be signed. This parameter is REQUIRED if the first option for the usage of this message is chosen. It SHALL be missing if the second option is used.
idPICC	CONDITIONAL An identifier of the eMRTD. In the specification of TA (see [BSI-TR-03110-1]) this identifier is denoted as ID_{PICC} . This parameter is REQUIRED if the second option is chosen. It SHALL be missing if the first option is used.
challengePICC	CONDITIONAL The challenge of the eMRTD. In the specification of TA (see [BSI-TR-03110-1]) this challenge is denoted as r_{PICC} . This parameter is REQUIRED if the second option is chosen. It SHALL be missing if the first option is used.

Parameter	Description
hashPK	CONDITIONAL The hash value over the ephemeral public key PK_{PCD} of the terminal. In the specification of TA (see [BSI-TR-03110-1]) this value is denoted as $COMP(PK_{PCD})$. This parameter is REQUIRED if the second option is chosen. It SHALL be missing if the first option is used.
auxPCD	CONDITIONAL Auxiliary data of the terminal. In the specification of TA (see [BSI-TR-03110-1]) this data is denoted as A_{PCD}. This parameter is REQUIRED if the second option is chosen and some auxiliary terminal data is part of the data to be signed. It SHALL be missing if the first option is used.
Output	
signature	CONDITIONAL The calculated signature. It is REQUIRED if a valid signature has been calculated. It SHALL be missing if no valid signature was calculated.
returnCode	REQUIRED The following return codes are possible • ok_signature_available • failure_CHR_unknown • failure_internal_error • failure_other_error • failure_syntax
returnCodeMessage	OPTIONAL Further information regarding the processing of this message.

Table 5.2 Input and output for the message GetTASignature

5.3 Terminal Authentication for DVs that forbid callbacks from CVCA

The following message is used as part of TA for DVs that cannot receive callbacks from CVCA:

GetDVCertificates (►Section 5.3.1)

5.3.1 GetDVCertificates

A DV must obtain certificates from a CVCA. In the default case this is facilitated with the messages GetCertificates and RequestCertificate. These messages often require asynchronous processing. If the DV allows no callbacks, the CVCA cannot deliver the results. In such cases the DV can use GetDVCertificates to acquire the results of the asynchronous processing.

The parameters of the message GetDVCertificates are given in ►Table 5.3.

Parameter	Description
Input	
holderMnemonic	REQUIRED Identifies the DV that wants to receive the certificates it previously requested from the CVCA.
Output	

Parameter	Description
certificateSeq	CONDITIONAL If the message is processed successfully and the returnCode is not ok_cert_not_finished, this parameter SHALL contain the certificates requested by the DV identified by holderMnemonic.
returnCode	REQUIRED The following return codes are possible • ok_cert_available • ok_cert_not_finished • failure_internal_error • failure_messageID_unknown • failure_other_error • failure_syntax
returnCodeMessage	OPTIONAL Further information regarding the processing of this message.

Table 5.3 Input and output for the message GetDVCertificates

5.4 Terminal Authentication for DVs that request foreign certificates

The following message is used as part of TA for DVs that request foreign certificates:

RequestForeignCertificate (►Section 5.4.1)

5.4.1 RequestForeignCertificate

This message is used by a DV for initiating the request of a new certificate for one of its keys from a CVCA in another country. This message is not sent to the foreign CVCA which is intended to generate the certificate, instead it is sent to the national Single Point of Contact (SPOC) of the country of the DV. This national SPOC verifies the request of the DV according to national regulations. If the request meets the national regulations it is forwarded to the SPOC of the other country. See [CSN-36-9791] for details.

The parameters of the message RequestForeignCertificate are given in ►Table 5.4.

Parameter	Description
Input	
callbackIndicator	REQUIRED See [BSI-TR-03129-1] section 2.2.1 "Common Parameters".
messageID	OPTIONAL See [BSI-TR-03129-1] section 2.2.1 "Common Parameters".
responseURL	OPTIONAL See [BSI-TR-03129-1] section 2.2.1 "Common Parameters"..
foreignCAR	REQUIRED This parameter contains the reference to the (expected) signature key of the foreign certification authority which also should be contained in the body of the certificate request contained in the parameter certReq.
certReq	REQUIRED This parameter contains the certificate request. It must be constructed according to [BSI-TR-03129-3] section 4.1.1 "Parameters".
Output	

Parameter	Description
certificateSeq	CONDITIONAL If the message is processed successfully and the returnCode is ok_cert_available, this parameter SHALL contain the certificates requested by the DV with certReq.
returnCode	REQUIRED The following return codes are possible • ok_cert_available • ok_request_forwarded • ok_reception_ack • failure_domain_parameters • failure_expired • failure_foreignCAR_unknown • failure_inner_signature • failure_internal_error • failure_not_forwarded • failure_outer_signature • failure_request_not_accepted • failure_request_not_accepted_foreign • failure_syntax
returnCodeMessage	OPTIONAL Further information regarding the processing of this message.

Table 5.4 Input and output for the message RequestForeignCertificate

6 Protocols for Digital Seals

6.1 Implementation as Web Services

It is RECOMMENDED that the protocol given in this chapter is realized as a Web Service using the SOAP messages described in the attached WSDL specifications.

6.1.1 Services Implemented by a TCC

A TCC SHALL implement a Web Service towards the role "DV" supporting the message `SendSignerList` (►Section 6.2.2).

A TCC SHALL implement a Web Service towards its connected readers supporting the following messages:

- `GetSignerList` (►Section 6.2.1)
- `GetSignerCertificate` (►Section 6.3.1)
- `GetDigitalSealVerification` (►Section 6.3.2)

6.1.2 Services Implemented by a DV

A DV SHALL implement a Web Services towards the role "NPKD" supporting the message `SendSignerList` (►Section 6.2.2).

A DV SHALL implement a Web Services towards the role "TCC" supporting the message `GetSignerList` (►Section 6.2.1).

6.1.3 Services Implemented by a CVCA or NPKD

A NPKD (or CVCA) SHALL implement a Web Services towards the role "DV" supporting the message `GetSignerList` (►Section 6.2.1).

6.2 Distribution of Signer Lists

The following messages are used for the distribution of signer certificates:

- `GetSignerList` (►Section 6.2.1)
- `SendSignerList` (►Section 6.2.2)

These messages enable the distribution of signer certificates to facilitate verifying signatures from digital seals and eMRTDs that do not contain signer certificates and instead contain references to them (see ►Chapter 7 for digital seals and ►Section 4.2 for eMRTDs, respectively).

The types of signer lists supported by the messages are given in ►Table 6.1. The formats of those signer lists are described in ►Chapter 9¹.

Signer List Type / OID	Description
<code>id-DocumentSignerList</code>	Signer certificates for eMRTDs
<code>id-euHealthSealSignerList</code>	Signer certificates for EU type digital seals on health documents
<code>id-euHealthSealCscsList</code>	CSCA signer certificates for EU type digital seals on health documents
<code>id-icaoHealthSealSignerList</code>	Signer certificates for ICAO type digital seals on health documents
<code>id-icaoHealthSealCscsList</code>	CSCA signer certificates for ICAO type digital seals on health documents
<code>id-sovereignSealSignerList</code>	Signer certificates for digital seals on sovereign documents.
<code>id-sovereignSealCscsList</code>	CSCA signer certificates for digital seals on sovereign documents.

Table 6.1 Types of signer lists

¹ There is no affiliation to the institutions ICAO or EU.

6.2.1 GetSignerList

The message `GetSignerList` is used to get a signed list of signer certificates. Its parameters are given in ▶Table 6.2.

Parameter	Description
Input	
<code>signerListType</code>	REQUIRED The type of the list requested. See ▶Table 6.1 for possible values.
Output	
<code>signerList</code>	CONDITIONAL The signer list of type <code>signerListType</code> . This parameter is REQUIRED if a signer list is available. If no signer list of type <code>signerListType</code> is available, this parameter SHALL be omitted. See ▶Chapter 9 for details regarding the format.
<code>returnCode</code>	REQUIRED The following return codes are possible: • <code>ok_list_available</code> • <code>failure_list_not_available</code> • <code>failure_list_type_unknown</code> • <code>failure_internal_error</code> • <code>failure_other_error</code> • <code>failure_syntax</code>
<code>returnCodeMessage</code>	OPTIONAL Further information regarding the processing of this message.

Table 6.2 Input and output for the message `GetSignerList`

6.2.2 SendSignerList

The message `SendSignerList` is used to send a signed list of signer certificates. Its parameters are given in ▶Table 6.3.

Parameter	Description
Input	
<code>signerList</code>	REQUIRED A signed list of certificates of type <code>signerListType</code> . See ▶Chapter 9 for details regarding the format.
<code>signerListType</code>	REQUIRED See ▶Table 6.1 for possible values.
Output	
<code>returnCode</code>	REQUIRED The following return codes are possible: • <code>ok_received_correctly</code> • <code>failure_internal_error</code> • <code>failure_other_error</code> • <code>failure_syntax</code>

Parameter	Description
returnCodeMessage	OPTIONAL Further information regarding the processing of this message.

Table 6.3 Input and output for the message SendSignerList

6.3 Digital Seal Verification for distributed terminals

The following messages are used to verify Digital Seals within a distributed terminal.

- GetSignerCertificate (►Section 6.3.1)
- GetDigitalSealVerification (►Section 6.3.2)

6.3.1 GetSignerCertificate

The message GetSignerCertificate is used to get a single signer certificate from the TCC. The IS can use this signer certificate to verify the signature of a digital seal. Its parameters are given in ►Table 6.4.

Parameter	Description
Input	
signerCertificateIdentifier	REQUIRED This parameter SHALL be provided as it is encoded in the digital seal. The source of the value of this parameter depends on the type of digital seal. Please refer to the following list: • VDS: signerIdentifierAndCertificateReference ►Table 7.2 • VDS-NC: certificateReference ►Table 7.15 • EUCWT: keyIdentifier ►Table 7.35 • IDB: certificateReference ►Table 7.44
Output	
signerCertificateInformation	CONDITIONAL It is REQUIRED if the message is processed successfully. It SHALL be omitted otherwise. The structure of the parameter is given in ►Table 6.5.
returnCode	REQUIRED The following return codes are possible: • ok_certificate_available • failure_certificate_not_available • failure_internal_error • failure_other_error • failure_syntax
returnCodeMessage	OPTIONAL Further information regarding the processing of this message.

Table 6.4 Input and output for the message GetSignerCertificate

Information about a certificate identified by signerCertificateIdentifier is given in ►Table 6.5.

Parameter	Description
certificate	REQUIRED The certificate that is identified by the parameter signerCertificateIdentifier as given in ►Table 6.4.

Parameter	Description
knownDefects	CONDITIONAL A list of defects for the given certificate known to the TCC. This parameter is REQUIRED if the TCC knows defects for the given certificate. It SHALL be omitted otherwise. The structure of a single entry is given in ▶Table 4.8.
unknownCriticalExtension	CONDITIONAL Unknown critical extension contained in the certificate. It is REQUIRED if such extensions exist. It SHALL be omitted otherwise.
listStatusInformation	CONDITIONAL A list of listStatus elements. The format of a listStatus is described in ▶Table 3.3. REQUIRED if there are lists present in the TCC, SHALL be omitted otherwise.

Table 6.5 signerCertificateInformation (signerCertificateInformationType) parameter definitions

6.3.2 GetDigitalSealVerification

The message GetDigitalSealVerification is used to get the result of a verification of a digital seal from the TCC. Its parameters are given in ▶Table 6.6.

Parameter	Description
Input	
digitalSealContent	REQUIRED The content of the digital seal encoded as binary.
MRZFirstLine	CONDITIONAL The first line of the Machine Readable Zone (MRZ). It is REQUIRED if the document has a two or three line MRZ. It SHALL be omitted otherwise.
MRZSecondLine	CONDITIONAL The second line of the MRZ. It is REQUIRED if the document has a two or three line MRZ. It SHALL be omitted otherwise.
MRZThirdLine	CONDITIONAL The third line of the MRZ. It is REQUIRED if the document has a three line MRZ. It SHALL be omitted otherwise.
returnCACertificate	REQUIRED Indicates whether the caller wants to receive the corresponding CA certificate.
returnSignerCertificate	REQUIRED Indicates whether the caller wants to receive the corresponding signer certificate.
Output	
digitalSealVerificationResult	CONDITIONAL This output is REQUIRED if the message was processed successfully. The structure of the parameter is given in ▶Table 6.7.

Parameter	Description
returnCode	REQUIRED The following return codes are possible: • ok_verification_done • failure_internal_error • failure_other_error • failure_syntax • failure_verification_not_possible
returnCodeMessage	OPTIONAL Further information regarding the processing of this message.

Table 6.6 Input and output for the message GetDigitalSealVerification

The result of a verification of a digital seal is given in digitalSealVerificationResult. The corresponding type definition is according to ▶Table 6.7.

Parameter	Description
trustStatus	REQUIRED The TCC's rating for the signerCertificate. The following values are possible: • certificate_ok • certificate_undetermined • certificate_invalid
certificateChainValidity	REQUIRED Information about the validity of the certificate chain. The structure of the parameter is given in ▶Table 4.5.
signerCertificateSignature	REQUIRED Result of the validation of the signature of the signerCertificate. The structure of the parameter is given in ▶Table 4.6.
knownDefects	CONDITIONAL A list of defects for the signerCertificate known to the TCC. This parameter is REQUIRED if the TCC knows defects for the signerCertificate. Otherwise, it SHALL be omitted. The structure of a single entry is given in ▶Table 4.8.

Parameter	Description
<code>parsedDigitalSeal</code>	REQUIRED This parameter has one of the following types, depending on the digital seal format (see ▶Chapter 7) detected by the TCC: • VDS The structure of this type is given in ▶Table 7.1 • VDSNC The structure of this type is given in ▶Table 7.11 • EUCWT The structure of this type is given in ▶Table 7.34 • IDB The structure of this type is given in ▶Table 7.42 • <code>digitalSealNotParsable</code> The raw binary content of the digital seal in base64 encoding.
<code>digitalSealContentDomain</code>	REQUIRED A digital seal contains information for a specific domain. The content of the parsed digital seal SHALL be categorized in one of the following domains: • <code>address_sticker</code> • <code>arrival_attestation</code> • <code>card_access_number</code> • <code>digital_travel_authorization</code> • <code>ef_card_access</code> • <code>emergency_travel_document</code> • <code>health_proof</code> • <code>machine_readable_zone</code> • <code>multiple</code> • <code>proof_of_recovery</code> • <code>proof_of_testing</code> • <code>proof_of_vaccination</code> • <code>residence_permit</code> • <code>social_insurance_card</code> • <code>travel_authorization</code> • <code>unknown</code> • <code>visa</code>
<code>signerCertificateDomain</code>	REQUIRED Each certificate is contained in a certificate domain represented by a certificate list similarly to the Master list. A known certificate SHALL be identified by the OID of its domain (cmp. ▶Table 6.1). If the certificate domain is unknown this value SHALL be: <code>certificate_domain_unknown</code>.

Parameter	Description
MRZvsDigitalSealCheck	CONDITIONAL The result of the comparison of the MRZ and information contained in the digital seal. It is REQUIRED if an MRZ was sent as input and SHALL be omitted otherwise. The result of this check SHALL be one of the following: • identical • not_checkable • not_identical
caCertificate	CONDITIONAL The corresponding CA certificate. It is REQUIRED if the CA certificate that was used to sign the signerCertificate is known to the TCC. It SHALL be omitted otherwise.
signerCertificate	CONDITIONAL The corresponding signer certificate. It is REQUIRED if the signer certificate that was used to sign the digital seal is known to the TCC. It SHALL be omitted otherwise.
unknownCriticalExtension	CONDITIONAL Unknown critical extensions contained in signerCertificate. It is REQUIRED if such extensions exist. It SHALL be omitted otherwise.
listStatusInformation	CONDITIONAL A list of listStatus elements. The format of a listStatus is described in ►Table 3.3. REQUIRED if there are lists present in the TCC, SHALL be omitted otherwise.

Table 6.7 digitalSealVerificationResult (digitalSealVerificationResultType) definition

7 Digital Seal Formats

Different institutions define their own formats for digital seals. The following formats are used in the context of this TR:

- VDS (see ▶Section 7.1 and [ICAO Doc 9303 p13])
- VDSNC (see ▶Section 7.2 and [ICAO TR VDS-NC])
- EUCWT (see ▶Section 7.3 and [EU DCC JSON 1.3.0])
- IDB (see ▶Section 7.4 and [ICAO TR IDB])

7.1 VDS

A Visible Digital Seal (VDS) is a type of digital seal specified by the International Civil Aviation Organization (ICAO) in [ICAO Doc 9303 p13].

The format of a VDS is described in ▶Table 7.1.

Parameter	Description
headerInformation	REQUIRED See [ICAO Doc 9303 p13]. The structure of this parameter is given in ▶Table 7.2.
message	REQUIRED See [ICAO Doc 9303 p13]. The structure of this parameter is given in ▶Table 7.3
signature	REQUIRED See [ICAO Doc 9303 p13].

Table 7.1 VDS (VDSType) definition

Header information contained in a VDS is transported in VDS via headerInformation. The parameter has the type VDS.headerType and its structure is given in ▶Table 7.2.

Parameter	Description
magicConstant	REQUIRED See [ICAO Doc 9303 p13].
version	REQUIRED See [ICAO Doc 9303 p13].
issuingCountry	REQUIRED See [ICAO Doc 9303 p13].
signerIdentifierAndCertificateReference	REQUIRED See [ICAO Doc 9303 p13].
documentIssueDate	REQUIRED See [ICAO Doc 9303 p13].
signatureCreationDate	REQUIRED See [ICAO Doc 9303 p13].
documentfeatureDefinitionReference	REQUIRED See [ICAO Doc 9303 p13].
documentTypeCategory	REQUIRED See [ICAO Doc 9303 p13].

Table 7.2 headerInformation (VDS.headerType) definition

The payload of a VDS is contained in `message`. That parameter has the type `VDS.messageType` and its structure is given in ▶Table 7.3.

Parameter	Description
<code>content</code>	REQUIRED This parameter has one of the following types, depending on the message format detected by the TCC: • <code>visa</code> (<code>VDS.messageVisaType</code>) See [ICAO Doc 9303 p7]. The structure of this type is given in ▶Table 7.4. • <code>etd</code> (<code>VDS.messageEmergencyTravelDocumentType</code>) See [ICAO Doc 9303 p8]. The structure of this type is given in ▶Table 7.5. • <code>aad</code> (<code>VDS.messageArrivalAttestationDocumentType</code>) See [BSI-TR-03137-1]. The structure of this type is given in ▶Table 7.6. • <code>sic</code> (<code>VDS.messageSocialInsuranceCardType</code>) See [BSI-TR-03137-1]. The structure of this type is given in ▶Table 7.7. • <code>rp</code> (<code>VDS.messageResidencePermitType</code>) See [BSI-TR-03137-1]. The structure of this type is given in ▶Table 7.8. • <code>rpss</code> (<code>VDS.messageResidencePermitSupplementarySheetType</code>) See [BSI-TR-03137-1]. The structure of this type is given in ▶Table 7.9. • <code>asgic</code> (<code>VDS.messageAddressStickerGermanIdentityCardType</code>) See [BSI-TR-03137-1]. The structure of this type is given in ▶Table 7.10. • <code>unspecified</code> (<code>VDS.messageUnknownType</code>) Content of an unknown message format in base64 encoding.
<code>status</code>	REQUIRED The following values are possible: • <code>ok</code> • <code>ok_unknown_feature_detected</code> • <code>failure_format_invalid</code> • <code>failure_format_not_recognized</code>

Table 7.3 message (`VDS.messageType`) definition

7.1.1 Visa

The use of VDS for visa is described in [ICAO Doc 9303 p7].

The parameter `visa` is of type `VDS.messageVisaType` whose structure is given in ▶Table 7.4.

Parameter	Description
<code>MRZ</code>	REQUIRED See [ICAO Doc 9303 p7].

Parameter	Description
numberOfEntries	CONDITIONAL See [ICAO Doc 9303 p7]. It is REQUIRED if it is present in the digital seal. It SHALL be omitted otherwise.
durationOfStay	REQUIRED See [ICAO Doc 9303 p7].
passportNumber	REQUIRED See [ICAO Doc 9303 p7].
visaType	CONDITIONAL See [ICAO Doc 9303 p7]. It is REQUIRED if it is present in the digital seal. It SHALL be omitted otherwise.
additionalFeature	CONDITIONAL See [ICAO Doc 9303 p7]. It is REQUIRED if it is present in the digital seal. It SHALL be omitted otherwise.

Table 7.4 visa (VDS.messageVisaType) definition

7.1.2 Emergency Travel Documents

The use of VDS for emergency travel documents is described in [ICAO Doc 9303 p8].

The parameter etd is of type VDS.messageEmergencyTravelDocumentType whose structure is given in ►Table 7.5.

Parameter	Description
MRZ	REQUIRED See [ICAO Doc 9303 p8].
additionalFeatures	CONDITIONAL See [ICAO Doc 9303 p8]. This parameter is REQUIRED if it is present in the digital seal. It SHALL be omitted otherwise.

Table 7.5 etd (VDS.messageEmergencyTravelDocumentType) definition

7.1.3 Arrival Attestation Documents

The use of VDS for arrival attestation documents is described in [BSI-TR-03137-1].

The parameter aad is of type VDS.messageArrivalAttestationDocumentType whose structure is given in ►Table 7.6.

Parameter	Description
MRZ	REQUIRED See [BSI-TR-03137-1].
azrNumber	REQUIRED See [BSI-TR-03137-1].

Table 7.6 aad (VDS.messageArrivalAttestationDocumentType) definition

7.1.4 Social Insurance Cards

The use of VDS for social insurance cards is described in [BSI-TR-03137-1].

The parameter sic is of type VDS.messageSocialInsuranceCardType whose structure is given in ►Table 7.7.

Parameter	Description
socialInsuranceNumber	REQUIRED See [BSI-TR-03137-1].
surname	REQUIRED See [BSI-TR-03137-1].
firstname	REQUIRED See [BSI-TR-03137-1].
nameAtBirth	CONDITIONAL See [BSI-TR-03137-1]. It is REQUIRED if it is present in the digital seal. It SHALL be omitted otherwise.

Table 7.7 sic (VDS.messageSocialInsuranceCardType) definition

7.1.5 Residence Permits

The use of VDS for residence permits is described in [BSI-TR-03137-1].

The parameter `rp` is of type `VDS.messageResidencePermitType` whose structure is given in ►Table 7.8.

Parameter	Description
MRZ	REQUIRED See [BSI-TR-03137-1].
passportNumber	REQUIRED See [BSI-TR-03137-1].

Table 7.8 rp (VDS.messageResidencePermitType) definition

7.1.6 Residence Permit Supplementary Sheets

The use of VDS for residence permit supplementary sheets is described in [BSI-TR-03137-1].

The parameter `rpss` is of type `VDS.messageResidencePermitSupplementarySheetType` whose structure is given in ►Table 7.9.

Parameter	Description
MRZ	REQUIRED See [BSI-TR-03137-1].
numberOfSupplementarySheet	REQUIRED See [BSI-TR-03137-1].

Table 7.9 rpss (VDS.messageResidencePermitSupplementarySheetType) definition

7.1.7 Address Sticker for German Identity Cards

The use of VDS for residence permit supplementary sheets is described in [BSI-TR-03137-1].

The parameter `asgc` is of type `VDS.messageAddressStickerGermanIdentityCardType` whose structure is given in ►Table 7.10.

Parameter	Description
documentNumber	REQUIRED See [BSI-TR-03137-1].
officialMunicipalityCodeNumber	REQUIRED See [BSI-TR-03137-1].

Parameter	Description
residentialAddress	REQUIRED See [BSI-TR-03137-1].

Table 7.10 asgic (VDS.messageAddressStickerGermanIdentityCardType) definition

7.2 VDS-NC

A Visible Digital Seal for Non-Constrained Environments (VDS-NC) is a type of digital seal specified by the ICAO in [ICAO TR VDS-NC].

The format of the parameter `VDSNC` is described in ▶Table 7.11.

Parameter	Description
data	REQUIRED See [ICAO TR VDS-NC]. The structure of the parameter is given in ▶Table 7.12.
signature	CONDITIONAL See [ICAO TR VDS-NC]. This parameter is REQUIRED if it is present in the digital seal. It SHALL be omitted otherwise. This parameter has one of the following types, depending on the signature detected by the TCC: • <code>signatureV1</code> Its structure is of type <code>(VDSNC . signatureV1Type)</code> as given in ▶Table 7.14. • <code>signatureV2</code> Its structure is of type <code>VDSNC . signatureV2Type</code> as given in ▶Table 7.15.

Table 7.11 VDSNC (VDSNC.VDSNCType) definition

The parameter `data` is of type `VDSNC . dataType` and its structure is given in ▶Table 7.12.

Parameter	Description
header	REQUIRED See [ICAO TR VDS-NC]. The structure of the parameter is given in ▶Table 7.13.

Parameter	Description
message	REQUIRED See [ICAO TR VDS-NC]. This parameter has one of the following types, depending on the message format detected by the TCC: • <code>digitalTravelAuthorization</code> Its structure is of type <code>VDSNC.digitalTravelAuthorizationMessageType</code> as given in ▶Table 7.16 • <code>proofOfVaccinationV1</code> Its structure is of type <code>VDSNC.proofOfVaccinationV1</code> as given in ▶Table 7.25 • <code>proofOfVaccinationV2</code> Its structure is of type <code>VDSNC.proofOfVaccinationV2</code> as given in ▶Table 7.26 • <code>proofOfTesting</code> Its structure is of type <code>VDSNC.proofOfTestingType</code> as given in ▶Table 7.19 • <code>unspecified (VDSNC.messageUnspecifiedType)</code> Raw content of an unknown format.

Table 7.12 data (`VDSNC.dataType`) definition

The parameter header is of type `VDSNC.headerType` and its structure is given in ▶Table 7.13.

Parameter	Description
type	REQUIRED See [ICAO TR VDS-NC].
version	REQUIRED See [ICAO TR VDS-NC].
issuingCountry	REQUIRED See [ICAO TR VDS-NC].

Table 7.13 header (`VDSNC.headerType`) definition

The parameter signatureV1 is of type `VDSNC.signatureV1Type` and its structure is given in ▶Table 7.14.

Parameter	Description
signatureAlgo	REQUIRED See [ICAO TR VDS-NC].
certificate	REQUIRED See [ICAO TR VDS-NC].
signatureValue	REQUIRED See [ICAO TR VDS-NC].

Table 7.14 signatureV1 (`VDSNC.signatureV1Type`) definition

The parameter signatureV2 is of type `VDSNC.signatureV2Type` and its structure is given in ▶Table 7.15.

Parameter	Description
signatureAlgo	REQUIRED See [ICAO TR VDS-NC].

Parameter	Description
certificate	CONDITIONAL See [ICAO TR VDS-NC]. It is REQUIRED if it is present in the digital seal. It SHALL be omitted otherwise.
certificateReference	CONDITIONAL See [ICAO TR VDS-NC]. It is REQUIRED if it is present in the digital seal. It SHALL be omitted otherwise.
signatureValue	REQUIRED See [ICAO TR VDS-NC].

Table 7.15 signatureV2 (VDSNC.signatureV2Type) definition

7.2.1 Digital Travel Authorization

Digital Travel Authorizations (DTAs) are defined in [ICAO TR DTAs].

The parameter `digitalTravelAuthorizationMessage` contains a DTAs. It is of type `VDSNC.digitalTravelAuthorizationMessageType`. Its structure is given in ▶Table 7.16.

Parameter	Description
dtaNumber	REQUIRED See [ICAO TR DTAs].
personInformation	REQUIRED The structure of the parameter is given in ▶Table 7.17.
digitalTravelAuthorization	REQUIRED The structure of the parameter is given in ▶Table 7.18.

Table 7.16 digitalTravelAuthorizationMessage (VDSNC.digitalTravelAuthorizationMessageType) definition

The parameter `personInformation` is of type `VDSNC.personInformationType`. Its structure is given in ▶Table 7.17.

Parameter	Description
name	REQUIRED See [ICAO TR DTAs].
passportNumber	REQUIRED See [ICAO TR DTAs].
dateOfBirth	REQUIRED See [ICAO TR DTAs].
nationality	CONDITIONAL See [ICAO TR DTAs]. It is REQUIRED if it is present in the digital seal. It SHALL be omitted otherwise.
sex	CONDITIONAL See [ICAO TR DTAs]. It is REQUIRED if it is present in the digital seal. It SHALL be omitted otherwise.

Table 7.17 personInformation (VDSNC.personInformationType) definition

The parameter `digitalTravelAuthorization` is of type `VDSNC.digitalTravelAuthorizationType`. Its structure is given in ▶Table 7.18.

Parameter	Description
placeOfIssue	REQUIRED See [ICAO TR DTAs].
validFrom	REQUIRED See [ICAO TR DTAs].
validUntil	REQUIRED See [ICAO TR DTAs].
durationOfStay	CONDITIONAL See [ICAO TR DTAs]. It is REQUIRED if it is present in the digital seal. It SHALL be omitted otherwise.
numberOfEntries	REQUIRED See [ICAO TR DTAs].
typeClassCategory	REQUIRED See [ICAO TR DTAs].
additionalInformation	CONDITIONAL See [ICAO TR DTAs]. It is REQUIRED if it is present in the digital seal. It SHALL be omitted otherwise.

Table 7.18 digitalTravelAuthorization (VDSNC.digitalTravelAuthorizationType) definition

7.2.2 Proof of Testing

The parameter proofOfTesting is of type VDSNC.proofOfTestingType. Its structure is given in ▶Table 7.19.

Parameter	Description
utci	CONDITIONAL See [ICAO TR VDS-NC]. It is REQUIRED if it is present in the digital seal and SHALL be omitted otherwise.
personalInformation	REQUIRED The structure of the parameter is given in ▶Table 7.20.
serviceProvider	REQUIRED The structure of the parameter is given in ▶Table 7.21.
dateTime	REQUIRED The structure of the parameter is given in ▶Table 7.23.
testResult	REQUIRED The structure of the parameter is given in ▶Table 7.24.
optionalData	CONDITIONAL See [ICAO TR VDS-NC]. It is REQUIRED if it is present in the digital seal. It SHALL be omitted otherwise.

Table 7.19 proofOfTesting (VDSNC.proofOfTestingType) definition

The parameter personalInformation is of type VDSNC.personalInformationType. Its structure is given in ▶Table 7.20.

Parameter	Description
name	REQUIRED See [ICAO TR VDS-NC].

Parameter	Description
dateOfBirth	REQUIRED See [ICAO TR VDS-NC].
documentType	REQUIRED See [ICAO TR VDS-NC].
documentNumber	REQUIRED See [ICAO TR VDS-NC].

Table 7.20 personalInformation (VDSNC.personalInformationType) definition

The parameter `serviceProvider` is of type `VDSNC.serviceProviderType`. Its structure is given in ►Table 7.21.

Parameter	Description
name	REQUIRED See [ICAO TR VDS-NC].
countryOfTest	REQUIRED See [ICAO TR VDS-NC].
contactDetails	REQUIRED The structure of the parameter is given in ►Table 7.22.

Table 7.21 serviceProvider (VDSNC.serviceProviderType) definition

The parameter `contactDetails` is of type `VDSNC.contactDetailsType`. Its structure is given in ►Table 7.22.

Parameter	Description
phoneNumber	REQUIRED See [ICAO TR VDS-NC].
eMail	REQUIRED See [ICAO TR VDS-NC].
address	REQUIRED See [ICAO TR VDS-NC].

Table 7.22 contactDetails (VDSNC.contactDetailsType) definition

The parameter `dateTime` is of type `VDSNC.dateTimeType`. Its structure is given in ►Table 7.23.

Parameter	Description
specimenCollection	REQUIRED See [ICAO TR VDS-NC].
reportIssuance	REQUIRED See [ICAO TR VDS-NC].

Table 7.23 dateTime (VDSNC.dateTimeType) definition

The parameter `testResult` is of type `VDSNC.testResultType`. Its structure is given in ►Table 7.24.

Parameter	Description
testConducted	REQUIRED See [ICAO TR VDS-NC].
result	REQUIRED See [ICAO TR VDS-NC].
method	OPTIONAL See [ICAO TR VDS-NC].

Table 7.24 testResult (VDSNC.testResultType) definition

7.2.3 Proof of Vaccination

There are two versions of a proof of vaccination:

- proofOfVaccinationV1 (►Table 7.25)
- proofOfVaccinationV2 (►Table 7.26)

The structure for version 1 of a proof of vaccination, i.e. proofOfVaccinationV1, is given in ►Table 7.25.

Parameter	Description
uvci	REQUIRED See [ICAO TR VDS-NC].
personIdentification	REQUIRED The structure of the parameter is given in ►Table 7.27.
vaccinationEvents	REQUIRED Contains one or more vaccinationEventV1 (vaccinationEventV1Type) (see ►Table 7.28) elements.

Table 7.25 proofOfVaccinationV1 (VDSNC.proofOfVaccinationV1Type) definition

The structure for version 2 of a proof of vaccination i.e. proofOfVaccinationV2, is given in ►Table 7.26.

Parameter	Description
uvci	REQUIRED See [ICAO TR VDS-NC].
certificateValidFrom	CONDITIONAL See [ICAO TR VDS-NC]. It is REQUIRED if it is present in the digital seal. It SHALL be omitted otherwise.
certificateValidUntil	CONDITIONAL See [ICAO TR VDS-NC]. It is REQUIRED if it is present in the digital seal. It SHALL be omitted otherwise.
personIdentification	REQUIRED The structure of the parameter is given in ►Table 7.27.
vaccinationEvents	REQUIRED Contains one or more vaccinationEventV2 (vaccinationEventV2Type) (see ►Table 7.29) elements.
optionalData	CONDITIONAL See [ICAO TR VDS-NC]. It is REQUIRED if it is present in the digital seal. It SHALL be omitted otherwise.

Table 7.26 proofOfVaccinationV2 (VDSNC.proofOfVaccinationV2Type) definition

The parameter personIdentification is of type VDSNC.personIdentificationType. Its structure is given in ►Table 7.27.

Parameter	Description
name	REQUIRED See [ICAO TR VDS-NC].
dateOfBirth	CONDITIONAL See [ICAO TR VDS-NC]. It is REQUIRED if it is present in the digital seal. It SHALL be omitted otherwise.

Parameter	Description
uniqueIdentifier	CONDITIONAL See [ICAO TR VDS-NC]. It is REQUIRED if it is present in the digital seal. It SHALL be omitted otherwise.
additionalIdentifier	CONDITIONAL See [ICAO TR VDS-NC]. It is REQUIRED if it is present in the digital seal. It SHALL be omitted otherwise.
sex	CONDITIONAL See [ICAO TR VDS-NC]. It is REQUIRED if it is present in the digital seal. It SHALL be omitted otherwise.

Table 7.27 personIdentificationType (VDSNC.personIdentificationType) definition

For each version of a proof of vaccination there is a corresponding vaccination event:

- vaccinationEventV1 and
- vaccinationEventV2.

The parameter vaccinationEventV1 is of type VDSNC.vaccinationEventV1Type. Its structure is given in ►Table 7.28.

Parameter	Description
vaccineProphylaxis	REQUIRED See [ICAO TR VDS-NC].
vaccineBrand	REQUIRED See [ICAO TR VDS-NC].
disease	OPTIONAL See [ICAO TR VDS-NC].
vaccinationDetails	REQUIRED The type of this parameter is VDSNC.vaccinationEventV1Type. Its structure is given in ►Table 7.30.

Table 7.28 vaccinationEventV1 (VDSNC.vaccinationEventV1Type) definition

The parameter vaccinationEventV2 is of type VDSNC.vaccinationEventV2Type. Its structure is given in ►Table 7.29.

Parameter	Description
vaccineProphylaxis	REQUIRED See [ICAO TR VDS-NC].
vaccineBrand	REQUIRED See [ICAO TR VDS-NC].
manufacturer	CONDITIONAL See [ICAO TR VDS-NC]. It is REQUIRED if it is present in the digital seal. It SHALL be omitted otherwise.
vaccineMarketAuthorizationHolder	CONDITIONAL See [ICAO TR VDS-NC]. It is REQUIRED if it is present in the digital seal. It SHALL be omitted otherwise.
disease	OPTIONAL See [ICAO TR VDS-NC].

Parameter	Description
vaccinationDetails	REQUIRED The type of this parameter is VDSNC.vaccinationEventV2Type. Its structure is given in ►Table 7.31.

Table 7.29 vaccinationEventV2 (VDSNC.vaccinationEventV2Type) definition

The vaccinationDetails of a version 1 vaccination event vaccinationEventV1 use the structure given in ►Table 7.30.

Parameter	Description
dateOfVaccination	REQUIRED See [ICAO TR VDS-NC].
doseNumber	REQUIRED See [ICAO TR VDS-NC].
countryOfVaccination	REQUIRED See [ICAO TR VDS-NC].
administeringCentre	REQUIRED See [ICAO TR VDS-NC].
batchNumber	REQUIRED See [ICAO TR VDS-NC].
dueDateNextDose	CONDITIONAL See [ICAO TR VDS-NC]. It is REQUIRED if it is present in the digital seal. It SHALL be omitted otherwise.

Table 7.30 vaccinationDetails (VDSNC.vaccinationDetailsV1Type) definition

The vaccinationDetails of a version 2 vaccination event vaccinationEventV2 use the structure given in ►Table 7.31.

Parameter	Description
dateOfVaccination	REQUIRED See [ICAO TR VDS-NC].
doseNumber	REQUIRED See [ICAO TR VDS-NC].
totalDoses	REQUIRED See [ICAO TR VDS-NC].
countryOfVaccination	REQUIRED See [ICAO TR VDS-NC].
administeringCentre	REQUIRED See [ICAO TR VDS-NC].
batchNumber	REQUIRED See [ICAO TR VDS-NC].
dueDateNextDose	CONDITIONAL See [ICAO TR VDS-NC]. It is REQUIRED if it is present in the digital seal. It SHALL be omitted otherwise.

Table 7.31 vaccinationDetails (VDSNC.vaccinationDetailsV2Type) definition

7.2.4 Proof of Recovery

The structure of a proof of recovery, i.e. `proofOfRecovery`, is according to type `VDSNC.proofOfRecoveryMessageType` as given in ▶Table 7.32.

Parameter	Description
<code>urci</code>	REQUIRED See [ICAO TR VDS-NC].
<code>certificateValidFrom</code>	CONDITIONAL See [ICAO TR VDS-NC]. It is REQUIRED if it is present in the digital seal. It SHALL be omitted otherwise.
<code>certificateValidUntil</code>	CONDITIONAL See [ICAO TR VDS-NC]. It is REQUIRED if it is present in the digital seal. It SHALL be omitted otherwise.
<code>personalInformation</code>	REQUIRED The structure of the parameter is given in ▶Table 7.20.
<code>testResult</code>	REQUIRED The structure of the parameter is given in ▶Table 7.33.
<code>optionalData</code>	CONDITIONAL See [ICAO TR VDS-NC]. It is REQUIRED if it is present in the digital seal. It SHALL be omitted otherwise.

Table 7.32 `proofOfRecovery` (`VDSNC.proofOfRecoveryMessageType`) definition

A test result, i.e. the parameter `testResult`, is given by the structure given in ▶Table 7.33.

Parameter	Description
<code>memberStateOfTest</code>	REQUIRED See [ICAO TR VDS-NC].
<code>dateOfFirstPositiveNAATTest-Result</code>	REQUIRED See [ICAO TR VDS-NC].

Table 7.33 `testResult` (`VDSNC.porTestResultType`) definition

7.3 EUCWT

The EU specifies digital seals containing health certificates (see [EU DCC Imp. Decision]). Those digital seals contain a CBOR Web Token (CWT) (see [RFC8392]) which is a cryptographically signed data structure (hereinafter called EUCWT).

The structure of a EUCWT is given in ▶Table 7.34.

Parameter	Description
<code>header</code>	REQUIRED The structure of the parameter is given in ▶Table 7.35.
<code>payload</code>	REQUIRED The structure of the parameter is given in ▶Table 7.36
<code>signature</code>	REQUIRED See [EU DCC Imp. Decision].

Table 7.34 `EUCWT.EUCWTType` definition

The parameter `header` has the type `EUCWT.headerType` and its structure is given in ▶Table 7.35.

Parameter	Description
signatureAlgorithm	REQUIRED See [EU DCC Imp. Decision].
keyIdentifier	REQUIRED See [EU DCC Imp. Decision].

Table 7.35 header (EUCWT.headerType) definition

The parameter payload has the type EUCWT.payloadType and its structure is given in ▶Table 7.36.

Parameter	Description
issuer	REQUIRED See [EU DCC Imp. Decision].
issuedAt	REQUIRED See [EU DCC Imp. Decision].
expirationTime	REQUIRED See [EU DCC Imp. Decision].
healthCertificate	REQUIRED See [EU DCC Imp. Decision]. This parameter has one of the following types, depending on the type detected by the TCC: • EUDCC Its structure is of type EUDCCType as given in ▶Table 7.37 • unspecified Raw content of an unknown format.

Table 7.36 payload(EUCWT.payloadType) definition

7.3.1 EU Digital COVID Certificate

The EUDCC is specified in ▶Table 7.37.

Parameter	Description
version	REQUIRED See [EU DCC JSON 1.3.0].
name	REQUIRED The structure of the parameter is given in ▶Table 7.38.
dateOfBirth	REQUIRED See [EU DCC JSON 1.3.0].

Parameter	Description
entry	REQUIRED This parameter has one of the following types, depending on the message entry format detected by the TCC: • vaccinationEntry See [EU DCC JSON 1.3.0]. This parameter is of type EUCWT . EUDCCEntryVaccinationType. Its structure is given in ▶Table 7.39. • testEntry See [EU DCC JSON 1.3.0]. This parameter is of type EUCWT . EUDCCEntryTestType. Its structure is given in ▶Table 7.40. • recoveryEntry See [EU DCC JSON 1.3.0]. This parameter is of type EUCWT . EUDCCEntryRecoveryType. Its structure is given in ▶Table 7.41.

Table 7.37 EUDCC (EUCWT.EUDGCType) definition

The parameter name has the type EUCWT . personNameType and its structure is given in ▶Table 7.38.

Parameter	Description
surnames	CONDITIONAL See [EU DCC JSON 1.3.0]. It is REQUIRED if it is present in the digital seal. It SHALL be omitted otherwise.
surnamesStandardised	CONDITIONAL See [EU DCC JSON 1.3.0]. It is REQUIRED if it is present in the digital seal. It SHALL be omitted otherwise.
fornames	CONDITIONAL See [EU DCC JSON 1.3.0]. It is REQUIRED if it is present in the digital seal. It SHALL be omitted otherwise.
fornamesStandardised	CONDITIONAL See [EU DCC JSON 1.3.0]. It is REQUIRED if it is present in the digital seal. It SHALL be omitted otherwise.

Table 7.38 name (EUCWT.personNameType) definition

7.3.1.1 Vaccination Entry

An EUDCC can contain information about a vaccination. The structure of the parameter `vaccinationEntry` is given in ▶Table 7.39.

Parameter	Description
disease	REQUIRED See [EU DCC JSON 1.3.0].
vaccineProphylaxis	REQUIRED See [EU DCC JSON 1.3.0].
vaccineProduct	REQUIRED See [EU DCC JSON 1.3.0].

Parameter	Description
vaccineMarketingAuthorizationHolder	REQUIRED See [EU DCC JSON 1.3.0].
doseNumber	REQUIRED See [EU DCC JSON 1.3.0].
totalDoses	REQUIRED See [EU DCC JSON 1.3.0].
dateOfVaccination	REQUIRED See [EU DCC JSON 1.3.0].
countryOfVaccination	REQUIRED See [EU DCC JSON 1.3.0].
certificateIssuer	REQUIRED See [EU DCC JSON 1.3.0].
certificateIdentifier	REQUIRED See [EU DCC JSON 1.3.0].

Table 7.39 vaccinationEntry (EUCWT.vaccinationEntryType) definition

7.3.1.2 Test Entry

An EUDCC can contain information about a test. The structure of the parameter testEntry is given in ►Table 7.40.

Parameter	Description
disease	REQUIRED See [EU DCC JSON 1.3.0].
testType	REQUIRED See [EU DCC JSON 1.3.0].
testName	CONDITIONAL See [EU DCC JSON 1.3.0]. It is REQUIRED if it is present in the digital seal. It SHALL be omitted otherwise.
testDeviceIdentifier	CONDITIONAL See [EU DCC JSON 1.3.0]. It is REQUIRED if it is present in the digital seal. It SHALL be omitted otherwise.
dateTimeTestSampleCollection	REQUIRED See [EU DCC JSON 1.3.0].
testResult	REQUIRED See [EU DCC JSON 1.3.0].
testCenter	CONDITIONAL See [EU DCC JSON 1.3.0]. It is REQUIRED if it is present in the digital seal. It SHALL be omitted otherwise.
countryOfTest	REQUIRED See [EU DCC JSON 1.3.0].
certificateIssuer	REQUIRED See [EU DCC JSON 1.3.0].

Parameter	Description
certificateIdentifier	REQUIRED See [EU DCC JSON 1.3.0].

Table 7.40 testEntry (EUCWT.testEntryType) definition

7.3.1.3 Recovery Entry

An EUDCC can contain information about a recovery entry. The structure of the parameter recoveryEntry is given in ▶Table 7.41.

Parameter	Description
disease	REQUIRED See [EU DCC JSON 1.3.0].
dateOfFirstPositiveNAATTest-Result	REQUIRED See [EU DCC JSON 1.3.0].
countryOfTest	REQUIRED See [EU DCC JSON 1.3.0].
certificateIssuer	REQUIRED See [EU DCC JSON 1.3.0].
certificateValidFrom	REQUIRED See [EU DCC JSON 1.3.0].
certificateValidUntil	REQUIRED See [EU DCC JSON 1.3.0].
certificateIdentifier	REQUIRED See [EU DCC JSON 1.3.0].

Table 7.41 recoveryEntry (EUCWT.recoveryEntryType) definition

7.4 IDB

An ICAO Datastructure for Barcode (IDB) is a type of digital seal format specified by the ICAO in [ICAO TR IDB]. It is a generic barcode format that can be used to encode signed and unsigned barcodes.

In the tables in this section the keyword **CONDITIONAL** in relation to a parameter **SHALL** be interpreted as follows: The parameter is **REQUIRED** if it is present in the digital seal. It **SHALL** be omitted otherwise.

The format of a IDB is described in ▶Table 7.42.

Parameter	Description
barcodeIdentifier	REQUIRED See [ICAO TR IDB].
barcodeFlag	CONDITIONAL See [ICAO TR IDB].
payload	CONDITIONAL See [ICAO TR IDB]. The structure of this parameter is given in ▶Table 7.43.

Table 7.42 IDB (IDBType) definition

The payload contained in a IDB is transported in an IDB via payload. The parameter is of type IDB.payloadType and its structure is given in ▶Table 7.43.

Parameter	Description
header	CONDITIONAL See [ICAO TR IDB]. The structure of this parameter is given in ▶Table 7.44.
message	CONDITIONAL See [ICAO TR IDB]. The structure of this parameter is given in ▶Table 7.45.
signerCertificate	CONDITIONAL See [ICAO TR IDB].
signature	CONDITIONAL See [ICAO TR IDB].

Table 7.43 payload (IDB.payloadType) definition

The header contained in a payload contains meta information on the corresponding barcode. The parameter is of type IDB.payloadHeaderType and its structure is given in ▶Table 7.44.

Parameter	Description
countryIdentifier	CONDITIONAL See [ICAO TR IDB].
signatureAlgorithm	CONDITIONAL See [ICAO TR IDB].
certificateReference	CONDITIONAL See [ICAO TR IDB].
signatureCreationDate	CONDITIONAL See [ICAO TR IDB].

Table 7.44 header (IDB.payloadHeaderType) definition

7.4.1 IDB message types

The message contained in a payload contains the data of one or more specific messages. The parameter is of type IDB.payloadMessageType and its structure is given in ▶Table 7.45.

Parameter	Description
Visa	CONDITIONAL See [ICAO TR IDB]. This message contains data of a visa. The message is encoded as given in [ICAO Doc 9303 p7], section 9.1.2. The parameter Visa is of type VDS.messageVisaType whose structure is given in ▶Table 7.46.
ETD	CONDITIONAL See [ICAO TR IDB]. This message contains data of a emergency travel document. The message is encoded as given in [ICAO Doc 9303 p8], section 6.1.2. The parameter ETD is of type IDB.proofOfTestingType whose structure is given in ▶Table 7.47.
PoT	CONDITIONAL See [ICAO TR IDB]. The parameter PoT is of type IDB.proofOfTestingType whose structure is given in ▶Table 7.48.

Parameter	Description
PoV	CONDITIONAL See [ICAO TR IDB]. The parameter PoV is of type IDB .proofOfVaccinationType whose structure is given in ▶Table 7.49.
PoR	CONDITIONAL See [ICAO TR IDB]. The parameter PoR is of type IDB .proofOfRecoveryType whose structure is given in ▶Table 7.52.
DTA	CONDITIONAL See [ICAO TR IDB]. The parameter DTA is of type IDB .digitalTravelAuthorizationType whose structure is given in ▶Table 7.53.
MRZTD1	CONDITIONAL See [ICAO TR IDB]. The parameter MRZTD1 is of type IDB .MRZTD1Type whose structure is given in ▶Table 7.54.
MRZTD3	CONDITIONAL See [ICAO TR IDB]. The parameter MRZTD3 is of type IDB .MRZTD3Type whose structure is given in ▶Table 7.55.
CAN	CONDITIONAL See [ICAO TR IDB]. The parameter CAN is of type IDB .CANType whose structure is given in ▶Table 7.56.
ECA	CONDITIONAL See [ICAO TR IDB]. The parameter ECA is of type IDB .ECAType whose structure is given in ▶Table 7.57.
Unparsable Message	CONDITIONAL Contains a message that could not be parsed. This parameter can occur more than once. It is of type IDB .unparsableMessage whose structure is given in ▶Table 7.58. It is REQUIRED if one or more unparsable messages are present in the digital seal. It SHALL be omitted otherwise.

Table 7.45 message (IDB.payloadMessageType) definition

7.4.1.1 Visa

The parameter Visa is of type IDB .VisaType whose structure is given in ▶Table 7.46.

Parameter	Description
MRZ	CONDITIONAL See [ICAO Doc 9303 p7].
numberOfEntries	CONDITIONAL See [ICAO Doc 9303 p7].
durationOfStay	CONDITIONAL See [ICAO Doc 9303 p7].
passportNumber	CONDITIONAL See [ICAO Doc 9303 p7].

Parameter	Description
visaType	CONDITIONAL See [ICAO Doc 9303 p7].
additionalFeature	CONDITIONAL See [ICAO Doc 9303 p7].

Table 7.46 Visa (IDB.visaType) definition

7.4.1.2 Emergency Travel Document

The parameter ETD is of type `VDS.messageEmergencyTravelDocumentType` whose structure is given in ▶Table 7.47.

Parameter	Description
MRZ	CONDITIONAL See [ICAO Doc 9303 p8].
additionalFeatures	CONDITIONAL See [ICAO Doc 9303 p8].

Table 7.47 ETD (IDB.emergencyTravelDocumentType) definition

7.4.1.3 Proof of Testing

The parameter PoT is of type `IDB.proofOfTestingType` and its structure is given in ▶Table 7.48.

Parameter	Description
uniqueTestCertificateIdentifier	CONDITIONAL See [ICAO TR IDB].
nameOfHolder	CONDITIONAL See [ICAO TR IDB].
dateOfBirth	CONDITIONAL See [ICAO TR IDB].
documentType	CONDITIONAL See [ICAO TR IDB].
documentNumber	CONDITIONAL See [ICAO TR IDB].
testingFacilityName	CONDITIONAL See [ICAO TR IDB].
countryOfTest	CONDITIONAL See [ICAO TR IDB].
testingFacilityPhone	CONDITIONAL See [ICAO TR IDB].
testingFacilityEmail	CONDITIONAL See [ICAO TR IDB].
testingFacilityAddress	CONDITIONAL See [ICAO TR IDB].
specimenCollectionDateTime	CONDITIONAL See [ICAO TR IDB].
reportIssuanceDateTime	CONDITIONAL See [ICAO TR IDB].

Parameter	Description
testConducted	CONDITIONAL See [ICAO TR IDB].
testResult	CONDITIONAL See [ICAO TR IDB].
samplingMethod	CONDITIONAL See [ICAO TR IDB].
optionalData	CONDITIONAL See [ICAO TR IDB].

Table 7.48 PoT (IDB.proofOfTestingType) definition

7.4.1.4 Proof of Vaccination

The parameter PoV is of type IDB.proofOfVaccinationType and its structure is given in ►Table 7.49.

Parameter	Description
uniqueVaccinationCertificateIdentifier	CONDITIONAL See [ICAO TR IDB].
certificateValidFrom	CONDITIONAL See [ICAO TR IDB].
certificateValidUntil	CONDITIONAL See [ICAO TR IDB].
nameOfHolder	CONDITIONAL See [ICAO TR IDB].
dateOfBirth	CONDITIONAL See [ICAO TR IDB].
sex	CONDITIONAL See [ICAO TR IDB].
uniqueIdentifier	CONDITIONAL See [ICAO TR IDB].
additionalIdentifier	CONDITIONAL See [ICAO TR IDB].
vaccinationEvents	CONDITIONAL This parameter is of type IDB.vaccinationEventsType. It contains one or more vaccinationEvent. The structure of vaccinationEvent is of type IDB.vaccinationEventType as given in ►Table 7.50
optionalData	CONDITIONAL See [ICAO TR IDB].

Table 7.49 PoV (IDB.proofOfVaccinationType) definition

The parameter vaccinationEvent is of type IDB.vaccinationEventType and its structure is given in ►Table 7.50.

Parameter	Description
vaccineOrProphylaxis	CONDITIONAL See [ICAO TR IDB].

Parameter	Description
vaccineBrand	CONDITIONAL See [ICAO TR IDB].
manufacturer	CONDITIONAL See [ICAO TR IDB].
vaccineMarketAuthorizationHolder	CONDITIONAL See [ICAO TR IDB].
diseaseTargeted	CONDITIONAL See [ICAO TR IDB].
vaccinationDetails	CONDITIONAL See [ICAO TR IDB]. This parameter is of type IDB.vaccinationDetailsType. It contains one or more vaccinationDetail. The structure of vaccinationDetail is of type IDB.vaccinationDetailType as given in ►Table 7.51

Table 7.50 vaccinationEvent (IDB.vaccinationEventType) definition

The parameter vaccinationDetail is of type IDB.vaccinationDetailType and its structure is given in ►Table 7.51.

Parameter	Description
dateOfVaccination	CONDITIONAL See [ICAO TR IDB].
doseNumber	CONDITIONAL See [ICAO TR IDB].
totalDoses	CONDITIONAL See [ICAO TR IDB].
countryOfVaccination	CONDITIONAL See [ICAO TR IDB].
administeringCentre	CONDITIONAL See [ICAO TR IDB].
batchNumber	CONDITIONAL See [ICAO TR IDB].
dueDateNextDose	CONDITIONAL See [ICAO TR IDB].

Table 7.51 vaccinationDetail (IDB.vaccinationDetailType) definition

7.4.1.5 Proof of Recovery

The parameter PoR is of type IDB.proofOfRecoveryType and its structure is given in ►Table 7.52.

Parameter	Description
uniqueRecoveryCertificateIdentifier	CONDITIONAL See [ICAO TR IDB].
certificateValidFrom	CONDITIONAL See [ICAO TR IDB].

Parameter	Description
certificateValidUntil	CONDITIONAL See [ICAO TR IDB].
nameOfHolder	CONDITIONAL See [ICAO TR IDB].
dateOfBirth	CONDITIONAL See [ICAO TR IDB].
documentType	CONDITIONAL See [ICAO TR IDB].
documentNumber	CONDITIONAL See [ICAO TR IDB].
memberStateOfTest	CONDITIONAL See [ICAO TR IDB].
dateOfFirstPositiveTestResult	CONDITIONAL See [ICAO TR IDB].
optionalData	CONDITIONAL See [ICAO TR IDB].

Table 7.52 PoR (IDB.proofOfRecoveryType) definition

7.4.1.6 Digital Travel Authorization

The parameter DTA is of type IDB.digitalTravelAuthorizationType and its structure is given in ►Table 7.53.

Parameter	Description
digitalTravelAuthorizationNumber	CONDITIONAL See [ICAO TR IDB].
nameOfHolder	CONDITIONAL See [ICAO TR IDB].
passportNumber	CONDITIONAL See [ICAO TR IDB].
dateOfBirth	CONDITIONAL See [ICAO TR IDB].
nationality	CONDITIONAL See [ICAO TR IDB].
sex	CONDITIONAL See [ICAO TR IDB].
placeOfIssue	CONDITIONAL See [ICAO TR IDB].
validFrom	CONDITIONAL See [ICAO TR IDB].
validUntil	CONDITIONAL See [ICAO TR IDB].
durationOfStay	CONDITIONAL See [ICAO TR IDB].

Parameter	Description
numberOfEntries	CONDITIONAL See [ICAO TR IDB].
typeClassCategory	CONDITIONAL See [ICAO TR IDB].
additionalInformation	CONDITIONAL See [ICAO TR IDB].

Table 7.53 DTA (IDB.digitalTravelAuthorizationType) definition

7.4.1.7 Machine Readable Zone (TD1)

The parameter MRZTD1 is of type IDB.MRZTD1Type and its structure is given in ►Table 7.54.

Parameter	Description
MRZTD1	CONDITIONAL See [ICAO TR IDB].

Table 7.54 MRZTD1 (IDB.MRZTD1Type) definition

7.4.1.8 Machine Readable Zone (TD3)

The parameter MRZTD3 is of type IDB.MRZTD3Type and its structure is given in ►Table 7.55.

Parameter	Description
MRZTD3	CONDITIONAL See [ICAO TR IDB].

Table 7.55 MRZTD3 (IDB.MRZTD3Type) definition

7.4.1.9 Card Access Number

The parameter CAN is of type IDB.CANType and its structure is given in ►Table 7.56.

Parameter	Description
CAN	CONDITIONAL See [ICAO TR IDB].

Table 7.56 CAN (IDB.CANType) definition

7.4.1.10 EF.CardAccess

The parameter ECA is of type IDB.ECAType and its structure is given in ►Table 7.57.

Parameter	Description
ECA	CONDITIONAL See [ICAO TR IDB].

Table 7.57 ECA (IDB.ECAType) definition

7.4.1.11 Unparsable Message

The parameter unparsableMessage is of type IDB.unparsableMessage and its structure is given in ►Table 7.58.

Parameter	Description
messageTypeTag	CONDITIONAL The tag of the unparsable message.

Parameter	Description
message	CONDITIONAL The complete base64-encoded unparsable message.

Table 7.58 unparsableMessage (IDB.unparsableMessage) definition

8 Defect Lists for ePassport Application and Digital Seals

A Defect is a production error affecting a large number of documents. The withdrawal of already issued documents is at best impractical and impossible if foreign documents are affected. Defects are identified by the signer certificate(s) used to produce defect documents or the CA certificate(s) used to sign the signer certificates..

Defect Lists (DFLs) are errata that inform about defects and provide corrigenda to fix errors if possible. DFLs are provided as Signed Data according to [RFC5652] and SHALL be using the profile and list content description as specified in [BSI-TR-03129-3].

This chapter defines DFLs for the handling of Defects in the context of official document checks making use of the ePassport Application on a chip or digital seals. Defects relating to the eID Application are defined in [BSI-TR-03129-3].

Defect handling procedures for IS are defined in [BSI-TR-03135-1].

8.1 Defect List Format

An inspection system compliant with this TR SHALL support version 1, version 2 and version 3 of the DFL format. A DFL SHALL be provided as Basic Encoding Rules (BER) encoded SignedData with content type DefectList identified by id-DefectList (see [BSI-TR-03129-3]):

```
bsi-de OBJECT IDENTIFIER ::= {
    itu-t(0) identified-organization(4) etsi(0)
    reserved(127) etsi-identified-organization(0) 7
}

id-DefectList OBJECT IDENTIFIER ::= { bsi-de applications(3) mrtid(1) 5 }
```

```
DefectList ::= SEQUENCE {
    version INTEGER {v1(0), v2(1), v3(2)},
    hashAlg OBJECT IDENTIFIER,
    defects SET OF Defect
}
```

Version 2 of the DFL format introduces an OPTIONAL description field. It can contain a human readable description of the class of a Defect and additional information and notes on the Defect or affected documents. A Defect is specified as given below:

```
Defect ::= SEQUENCE {
    signerIdentifier SignerIdentifier OPTIONAL,           -- present in v1, MUST NOT be
    present in v3                                         -- present in v1, MUST NOT be
    certificateHash OCTET STRING OPTIONAL,              -- MUST NOT be present in v3
    signerCertificateReference SignerCertificateReference OPTIONAL, -- MUST be present in v3 if
    caCertificateReference is not present                -- MUST be present in v3 if
    caCertificateReference CaCertificateReference OPTIONAL, -- MUST be present in v3 if
    signerCertificateReference is not present            -- MUST be present in v3 if
    knownDefects SET OF VersionedKnownDefect,
    description UTF8String OPTIONAL                    -- MAY only be present in v2 or
    v3
}
```

The type SignerIdentifier is defined in [RFC5652] as follows:

```
SignerIdentifier ::= CHOICE {
    issuerAndSerialNumber IssuerAndSerialNumber,
    subjectKeyIdentifier [0] SubjectKeyIdentifier
}
```

The type CertificateReference is defined as follows:

```
SignerCertificateReference ::= SEQUENCE {
    issuerAndSerialNumber [0] IssuerAndSerialNumber OPTIONAL, -- REQUIRED if available,
```

```

Definition
subjectKeyIdentifier [1] SubjectKeyIdentifier OPTIONAL, -- see SignerIdentifier for the
-- REQUIRED if available,
-- see SignerIdentifier for the
Definition
signerReference [2] UTF8String OPTIONAL, -- REQUIRED if available,
-- see explanation below
certificateHash [3] OCTET STRING OPTIONAL -- REQUIRED if available
}

```

The element `signerReference` is an alphanumerical string that refers to a signer certificate. See `signerIdentifierAndCertificateReference` in ▶Table 7.2, `certificateReference` in ▶Table 7.15, `keyIdentifier` in ▶Table 7.35 and `certificateReference` in ▶Table 7.44.

```

CaCertificateReference ::= SEQUENCE {
  issuer [0] Name OPTIONAL, -- REQUIRED if available,
  authorityKeyIdentifier [1] AuthorityKeyIdentifier OPTIONAL, -- REQUIRED if available
}

```

The type `Name` is defined in [RFC5652]. The type `AuthorityKeyIdentifier` is defined in [RFC5280].

A `VersionedKnownDefect` SHALL be either a `knownDefect` OR `knownDefectV2`:

```

VersionedKnownDefect ::= CHOICE {
  knownDefect KnownDefect, -- use this if version is v1
  knownDefectV2 [0] KnownDefectV2 -- use this if version is v2 or v3
}

```

The `OPTIONAL` description field in `KnownDefectV2` can contain a description of the known defect with additional technical details:

```

KnownDefect ::= SEQUENCE {
  defectType OBJECT IDENTIFIER,
  parameters ANY DEFINED BY defectType OPTIONAL
}

KnownDefectV2 ::= SEQUENCE {
  defectType OBJECT IDENTIFIER,
  parameters [0] ANY DEFINED BY defectType OPTIONAL,
  description [1] UTF8String OPTIONAL
}

```

8.1.1 Identification of Certificates

A defect can contain one of two types of references. One is a reference to a signer certificate and the other is a reference to a CA certificate.

A defect SHALL be applied for a document containing a specific signer certificate if the defect references either this signer certificate itself or the corresponding CA certificate.

There are four possible ways to reference a signer certificate in a DFL:

1. the hash of the certificate. The hash function to be used is indicated in `DefectList`.
2. the `signerReference` of the certificate
3. the Subject Key Identifier (SKI) of the certificate
4. the distinguished name of the CA certificate (issuer) and the serial number of the certificate

The preference of the identifiers when matching a defect to a signer certificate SHOULD follow the listing above. The first list entry has the highest preference and the last list entry has the lowest preference. A match or mismatch in an identifier with higher preference SHALL overrule an identifier with lower preference.

There are two possible ways to reference a CA certificate in a DFL:

1. the authority key identifier that is used by signer certificates to reference the CA certificate
2. the distinguished name (issuer) of the CA certificate

The preference of the identifiers when matching a defect to a signer certificate SHOULD follow the listing above. The first list entry has the highest preferences and the last list entry has the lowest preference. A match or mismatch in an identifier with higher preference SHALL overrule an identifier with lower preference.

8.2 Defect List Signer

A defect list SHALL be signed by a DFL Signer. A DFL Signer certificate SHOULD include the following OID inside the extendedKeyUsage extension:

```
id-electronicDefectListSigningKey OBJECT IDENTIFIER ::= {
  bsi-de applications(3) id-npkd(11) id-defectExchange(2) id-electronicDefect(1) 2
}
```

This requirement will become mandatory in future versions of this TR.

Any system that is processing a DFL with the intent to directly or indirectly apply defects from that DFL SHALL check whether the DFL has a valid signature from a DFL Signer by using a DFL Signer certificate. If the signature is invalid, the system SHALL NOT use that DFL. If the signature is not from a DFL Signer, the system SHALL NOT use that DFL.

8.3 Defect Categories

There are three categories of defects:

- Authentication Defects
- Application Defects
- Document Defects

8.3.1 Authentication Defects

This section describes defects related to Passive Authentication, Chip Authentication and Active Authentication.

This category of Defects is indicated by the following object identifier:

```
id-AuthDefect OBJECT IDENTIFIER ::= {id-DefectList 1}
```

8.3.1.1 Signer Certificate Revoked

The private key of the signer certificate might be compromised (e.g. revoked by a Certificate Revocation List (CRL)).

Note that this defect is equivalent to an issued CRL containing the signer certificate. This type of defect can be used to revoke foreign signer certificates independently from a CRL.

This type of defect is indicated by the following object identifier:

```
id-CertRevoked OBJECT IDENTIFIER ::= {id-AuthDefect 1}
```

The following parameter MUST be present:

```
StatusCode ::= ENUMERATED {
  noIndication(0),           -- no details given
  onHold(1),                 -- revocation under investigation
  testing(2),                -- the certificate has been used for testing
  revokedByIssuer(3),        -- the Issuer has revoked the certificate by CRL
  revokedDLS(4),             -- the Defect List Signer has revoked the certificate
  certInadequate(5),         -- the certificate is inadequate
  proprietary(32)            -- status codes >=32 for internal use
}
```

8.3.1.2 Signer Certificate Malformed

The signer certificate might be malformed and therefore might not be decoded correctly. Instead a replacement certificate is provided. *Note that this certificate is not necessarily signed by the original issuer. Replacement certificates MAY be self-signed.*

This type of defect is indicated by the following object identifier:

```
id-CertReplaced OBJECT IDENTIFIER ::= {id-AuthDefect 2}
```

The following parameter MUST be present:

```
ReplacementCertificate ::= OCTET STRING
```

The parameter ReplacementCertificate contains a certificate (Certificate, cf. [RFC5280]) as replacement for the signer certificate.

8.3.1.3 Chip Authentication Private Keys Compromised

The Chip Authentication Private Keys might have been compromised (e.g. due to an error in the key generation algorithm).

This type of defect is indicated by the following object identifier:

```
id-ChipAuthKeyRevoked OBJECT IDENTIFIER ::= {id-AuthDefect 3}
```

There are no parameters provided.

8.3.1.4 Active Authentication Private Keys Compromised

The Active Authentication Private Keys might have been compromised (e.g. due to an error in the key generation algorithm).

This type of defect is indicated by the following object identifier:

```
id-ActiveAuthKeyRevoked OBJECT IDENTIFIER ::= {id-AuthDefect 4}
```

There are no parameters provided.

8.3.1.5 Authentication Protocol Failure

Active Authentication or Chip Authentication might be performed incorrectly.

This type of defect is indicated by the following object identifier:

```
id-AuthenticationProtocolFailure OBJECT IDENTIFIER ::= {id-AuthDefect 5}
```

The following parameter MUST be present:

```
StatusCode ::= ENUMERATED {
  CA_failure(0),           -- Chip Authentication is known to fail
  AA_failure(1),           -- Active Authentication is known to fail
  CA_AA_failure(2),        -- Chip Authentication and Active Authentication are known to fail
  PACE_CAM_failure(3),     -- PACE-CAM is known to fail
  CA_PACE_CAM_failure(4),  -- Chip Authentication and PACE-CAM are known to fail
  AA_PACE_CAM_failure(5),  -- Active Authentication and PACE-CAM are known to fail
  CA_AA_PACE_CAM_failure(6) -- Chip Authentication, Active Authentication and PACE-CAM are known
  to fail
}
```

8.3.1.6 Validity Period Incorrect

The validity period of the signer certificate or CSCA certificate might be incorrect.

This type of defect is indicated by the following object identifier:

```
id-ValidityPeriodIncorrect OBJECT IDENTIFIER ::= {id-AuthDefect 6}
```

The following parameter **MUST** be present:

```
StatusCode ::= ENUMERATED {
  CSCA_validity(0),      -- CSCA certificate validity is incorrect
  DS_validity(1),        -- signer certificate validity is incorrect
  CSCA_DS_validity(2)    -- Validity of both the signer and the CSCA certificate is
    incorrect
}
```

8.3.2 Application Defects (for the ePassport Application)

The following sections describe Defects related to the personalisation of the ePassport application.

This category of Defects is indicated by the following object identifier:

```
id-ePassportDefect OBJECT IDENTIFIER ::= {id-DefectList 2}
```

8.3.2.1 Data Group Malformed

The indicated data groups might be incorrectly encoded.

This type of defect is indicated by the following object identifier:

```
id-ePassportDGMalformed OBJECT IDENTIFIER ::= {id-ePassportDefect 1}
```

The following parameter **MUST** be present:

```
MalformedDGs ::= SET OF INTEGER      -- DGs as integer
```

The parameter `MalformedDGs` lists the malformed datagroups.

8.3.2.2 Document Security Object Malformed

The validation of the Document Security Object might fail (e.g. signature incorrect).

This type of defect is indicated by the following object identifier:

```
id-SODInvalid OBJECT IDENTIFIER ::= {id-ePassportDefect 2}
```

The following parameter **MAY** be present:

```
SodFaultStatusCode ::= ENUMERATED {
  noIndication(0),      -- no details given
  sodSignatureFailure(1), -- Failure in SOD's signature
  sodHashInvalid(2),    -- sodHashInvalid
  sodEncodingFailure(3), -- wrong encoding / encoding error
  sodDGHashesInvalid(4), -- wrong/invalid DG hashes (or encoding)
  proprietary(32)       -- >=32 codes used for internal purposes
}
```

8.3.2.3 COM and SOD Discrepancy

The list of present datagroups contained in EF.COM and/or EF.SOD might not be correct.

This type of defect is indicated by the following object identifier:

```
id-COMSODDiscrepancy OBJECT IDENTIFIER ::= {id-ePassportDefect 3}
```

The following parameter **MAY** be present:

```
DataGroupDiscrepancy ::= SEQUENCE {
  listedInCOM SET OF INTEGER, -- data groups as defined through COM
  listedInSOD SET OF INTEGER  -- data groups as defined through SOD
}
```

```
}

```

The parameter `dataGroupDiscrepancy` defines which datagroups are present in the EF.COM and EF.SOD. If it is not present, the discrepancy is interpreted as a generic one.

8.3.2.4 Wrong Signer Identifier

The signer identifier inside the EF.SOD or digital seal might be wrong, hence there might be no matching signer certificate in the security object or signer list.

This type of defect is indicated by the following object identifier:

```
id-sodWrongSignerIdentifier OBJECT IDENTIFIER ::= {id-ePassportDefect 4}

```

The following parameter **MUST** be provided:

```
correctedSignerIdentifier ::= SignerIdentifier

```

The parameter `correctedSignerIdentifier` contains the correct signer identifier.

8.3.2.5 Issuing Country Defect (DEPRECATED)

This defect is DEPRECATED. Please use `Distinguished Name Defect` instead.

The country element of the issuer's distinguished name inside the signer certificate might be missing or incorrect. (e.g. using an ICAO 3-letter code instead of an ICAO 2-letter code, see [ICAO Doc 9303 p3], or using the wrong value).

This type of defect is indicated by the following object identifier:

```
id-IssuingCountryDefect OBJECT IDENTIFIER ::= {id-ePassportDefect 5}

```

The following parameter **MUST** be provided:

```
correctedIssuingCountry ::= PrintableString

```

8.3.2.6 Distinguished Name Defect

A part of the subject's or issuer's distinguished name inside the certificate might be missing or incorrect.

An example is using an ICAO 3-letter code instead of an ICAO 2-letter code (see [ICAO Doc 9303 p3]) inside the country name part of the issuer's distinguished name.

This type of defect is indicated by the following object identifier:

```
id-DistinguishedNameDefect OBJECT IDENTIFIER ::= {id-ePassportDefect 6}

```

The following parameter **MUST** be provided:

```
DistinguishedNameDefectInformation ::= SET SIZE (1..MAX) OF DistinguishedNameDefectErroneousPart

```

```
DistinguishedNameDefectErroneousPart ::= SEQUENCE {
    type          DistinguishedNameDefectType
    part          AttributeType
    incorrectValue [0] AttributeValue OPTIONAL -- MUST NOT be present if the value is missing,
    otherwise MUST be present
    correctedValue [1] AttributeValue
}

```

```
DistinguishedNameDefectType ::= ENUMERATED {
    Issuer(0),
    Subject(1)
}

```

The types `AttributeType` and `AttributeValue` are defined in [RFC5280].

8.3.3 General Document Defects

The following sections describe Defects related to the document in general.

This category of Defects is indicated by the following object identifier:

```
id-DocumentDefect OBJECT IDENTIFIER ::= {id-DefectList 4}
```

If a document is affected by a general defect (i.e. the defect is contained in the sub-tree of `id-DocumentDefect`) with unknown interpretation, the electronic part of the document **SHOULD NOT** be used for document checks, i.e. in this case, the data can not be verified.

8.3.3.1 Card Security Object Malformed

The Card Security Object might be incorrectly encoded. A corrected Card Security Object (see [BSI-TR-03110-3]) is provided and **SHOULD** be used.

This type of defect is indicated by the following object identifier:

```
id-CardSecurityMalformed OBJECT IDENTIFIER ::= {id-DocumentDefect 1}
```

The following parameter **MUST** be provided:

```
correctedCardSecurityObject ::= SignedData
```

8.3.3.2 Chip Security Object Malformed

The Chip Security Object might be incorrectly encoded. The Card Security Object **SHOULD** be used instead.

This type of defect is indicated by the following object identifier:

```
id-ChipSecurityMalformed OBJECT IDENTIFIER ::= {id-DocumentDefect 2}
```

There are no parameters provided.

8.3.3.3 Powerdown Required

The chip might deny multiple successive authentications using the General Authentication Procedure. Either the reader **SHALL** powerdown the chip or the chip/document **SHALL** be removed from the reader in between two authentications.

This type of defect is indicated by the following object identifier:

```
id-PowerDownReq OBJECT IDENTIFIER ::= {id-DocumentDefect 3}
```

There are no parameters provided.

8.3.3.4 Document Signer Certificate incorrectly encoded or malformed (DEPRECATED)

This defect is **DEPRECATED**. Please use `Certificate incorrectly encoded or malformed` instead.

The DS certificate might be incorrectly encoded or malformed and might be unusable. Compared to the Defect `id-CertReplaced` there is no replacement certificate provided. Additional information about this Defect is provided with the parameter `DSMalformedInformation`.

This type of defect is indicated by the following object identifier:

```
id-DSMalformed OBJECT IDENTIFIER ::= {id-DocumentDefect 4}
```

The following parameter **MUST** be provided:

```
DsMalformedInformation ::= ENUMERATED {
  noIndication(0),           -- no details given
  unknownCryptoAlg(1),       -- unknown cryptographic algorithm
  encodingFailure(2),        -- wrong encoding / encoding error
  proprietary(32)            -- codes >=32 can be used for internal purposes
}
```

```
}

```

8.3.3.5 Certificate incorrectly encoded or malformed

The certificate might be incorrectly encoded or malformed and might be unusable. Compared to the Defect `id-CertReplaced` there is no replacement certificate provided. Additional information about this Defect is provided with the parameter `CertificateMalformedInformation`.

This type of defect is indicated by the following object identifier:

```
id-CertificateMalformed OBJECT IDENTIFIER ::= {id-DocumentDefect 6}

```

The following parameter **MUST** be provided:

```
CertificateMalformedInformation ::= SEQUENCE {
    statusCode      CertificateMalformedStatusCode,
    certificateType  CertificateMalformedCertificateType
}

```

```
CertificateMalformedStatusCode ::= ENUMERATED {
    noIndication(0),          -- no details given
    unknownCryptoAlg(1),      -- unknown cryptographic algorithm
    encodingFailure(2),       -- wrong encoding / encoding error
    proprietary(32)           -- codes >=32 can be used for internal purposes,
                                -- all other unused codes are RFU
}

```

```
CertificateMalformedCertificateType ::= ENUMERATED {
    CountrySigning(0),        -- Root or Link Certificate of a Country Signing CA
    Signer(1)                 -- Document Signer, Barcode Signer or LDS2 Signer
}

```


9 Signed Data Profiles for Signer Lists

Signer Lists are provided as Signed Data according to [RFC5652] and SHALL be using the profile as defined in [BSI-TR-03129-3 Annex]. The value eContentType SHALL be determined according to ▶Table 9.1¹.

Signer List	eContentType / OID	Certificate Type
Document Signer List	id-DocumentSignerList	Document Signer
EU Health Seal Signer List	id-euHealthSealSignerList	Seal Signer
EU Health Seal CSCA List	id-euHealthSealCscalList	CSCA / Root
ICAO Health Seal Signer List	id-icaoHealthSealSignerList	Seal Signer
ICAO Health Seal CSCA List	id-icaoHealthSealCscalList	CSCA / Root
Sovereign Seal Signer List	id-sovereignSealSignerList	Seal Signer
Sovereign Seal CSCA List	id-sovereignSealCscalList	CSCA / Root

Table 9.1 eContentType of signer lists

Each list SHALL be provided as BER encoded Signed Data, i.e. each list is a signed set of x.509 certificates in BER encoding.

```
IMPORTS
-- Imports from RFC 5280 [PROFILE], Appendix A.1
Certificate
FROM PKIX1Explicit88 { iso(1) identified-organization(3)
    dod(6) internet(1) security(5) mechanisms(5) pkix(7)
    id-mod(0) id-pkix1-explicit(18)
}
```

Each list is defined in the bsi-de OID branch:

```
bsi-de OBJECT IDENTIFIER ::= {
    itu-t(0) identified-organization(4) etsi(0)
    reserved(127) etsi-identified-organization(0) 7
}
```

9.1 Document Signer List Format

A Document Signer List is defined as follows:

```
id-DocumentSignerList OBJECT IDENTIFIER ::= { bsi-de applications(3) mrttd(1) 6 }
```

```
documentSignerListVersion ::= INTEGER {v0(0)}
```

```
documentSignerList ::= SEQUENCE {
    version documentSignerListVersion,
    signerCertList SET OF Certificate
}
```

9.2 EU Health Seal Signer List Format

A EU Health Seal Signer List is defined as follows:

```
id-euHealthSealSignerList OBJECT IDENTIFIER ::= { bsi-de applications(3) id-npkd(11) id-
certificateExchange(1) id-euHealthSeal(2) 1 }
```

```
euHealthSealSignerListVersion ::= INTEGER {v0(0)}
```

¹ There is no affiliation to the institutions ICAO or EU.

```
euHealthSealSignerList ::= SEQUENCE {
    version euHealthSealSignerListVersion,
    signerCertList SET OF Certificate
}
```

9.3 EU Health Seal CSCA List Format

A EU Health Seal CSCA List is defined as follows:

```
id-euHealthSealCscalList OBJECT IDENTIFIER ::= { bsi-de applications(3) id-npkd(11) id-
certificateExchange(1) id-euHealthSeal(2) 3 }
```

```
euHealthSealCscalListVersion ::= INTEGER {v0(0)}
```

```
euHealthSealCscalList ::= SEQUENCE {
    version euHealthSealCscalListVersion,
    signerCertList SET OF Certificate
}
```

9.4 ICAO Health Seal Signer List Format

A ICAO Health Seal Signer List is defined as follows:

```
id-icaoHealthSealSignerList OBJECT IDENTIFIER ::= { bsi-de applications(3) id-npkd(11) id-
certificateExchange(1) id-icaoHealthSeal(3) 1 }
```

```
icaoHealthSealSignerListVersion ::= INTEGER {v0(0)}
```

```
icaoHealthSealSignerList ::= SEQUENCE {
    version icaoHealthSealSignerListVersion,
    signerCertList SET OF Certificate
}
```

9.5 ICAO Health Seal CSCA List Format

A ICAO Health Seal CSCA List is defined as follows:

```
id-icaoHealthSealCscalList OBJECT IDENTIFIER ::= { bsi-de applications(3) id-npkd(11) id-
certificateExchange(1) id-icaoHealthSeal(3) 3 }
```

```
icaoHealthSealCscalListVersion ::= INTEGER {v0(0)}
```

```
icaoHealthSealCscalList ::= SEQUENCE {
    version icaoHealthSealCscalListVersion,
    signerCertList SET OF Certificate
}
```

9.6 Sovereign Seal Signer List Format

A Sovereign Seal Signer List is defined as follows:

```
id-sovereignSealSignerList OBJECT IDENTIFIER ::= { bsi-de applications(3) id-npkd(11) id-
certificateExchange(1) id-sovereignSeal(1) 1 }
```

```
sovereignSealSignerListVersion ::= INTEGER {v0(0)}
```

```
sovereignSealSignerList ::= SEQUENCE {
    version sovereignSealSignerListVersion,
    signerCertList SET OF Certificate
}
```

```
}

```

9.7 Sovereign Seal CSCA List Format

A Sovereign Seal CSCA List is defined as follows:

```
id-sovereignSealCscalList OBJECT IDENTIFIER ::= { bsi-de applications(3) id-npkd(11) id-
certificateExchange(1) id-sovereignSeal(1) 3 }
```

```
sovereignSealCscalListVersion ::= INTEGER {v0(0)}
```

```
sovereignSealCscalList ::= SEQUENCE {
    version sovereignSealCscalListVersion,
    signerCertList SET OF Certificate
}
```

9.8 Signer List Signer Certificates

A Signer List Signer certificate SHALL include an OID in the `extendedKeyUsage` extension for every type of Signer List that the Signer List Signer shall be entitled to sign. The OIDs to use are given in ▶Table 9.2. A Signer List Signer certificate MAY include more than one of those OIDs inside its `extendedKeyUsage` extension to allow the Signer List Signer to use one private key to sign different signer list types.

Signer List Type	OID
Document Signer List	<code>id-documentSignerListSigningKey OBJECT IDENTIFIER ::= { bsi-de applications(3) id-npkd(11) id-certificateExchange(1) id-documentSigner(4) 2 }</code>
EU Health Seal Signer List	<code>id-euHealthSealSignerListSigningKey OBJECT IDENTIFIER ::= { bsi-de applications(3) id-npkd(11) id-certificateExchange(1) id-euHealthSeal(2) 2 }</code>
EU Health Seal CSCA List	<code>id-euHealthSealCscalListSigningKey OBJECT IDENTIFIER ::= { bsi-de applications(3) id-npkd(11) id-certificateExchange(1) id-euHealthSeal(2) 4 }</code>
ICAO Health Seal Signer List	<code>id-icaoHealthSealSignerListSigningKey OBJECT IDENTIFIER ::= { bsi-de applications(3) id-npkd(11) id-certificateExchange(1) id-icaoHealthSeal(3) 2 }</code>
ICAO Health Seal CSCA List	<code>id-icaoHealthSealCscalListSigningKey OBJECT IDENTIFIER ::= { bsi-de applications(3) id-npkd(11) id-certificateExchange(1) id-icaoHealthSeal(3) 4 }</code>
Sovereign Seal Signer List	<code>id-sovereignSealSignerListSigningKey OBJECT IDENTIFIER ::= { bsi-de applications(3) id-npkd(11) id-certificateExchange(1) id-sovereignSeal(1) 2 }</code>
Sovereign Seal CSCA List	<code>id-sovereignSealCscalListSigningKey OBJECT IDENTIFIER ::= { bsi-de applications(3) id-npkd(11) id-certificateExchange(1) id-sovereignSeal(1) 4 }</code>

Table 9.2 `extendedKeyUsage` of signer lists

A Signer List Signer Certificate SHALL include the private key usage period extension. Both parameters, i.e. `notBefore` and `notAfter`, SHALL be present. The private key usage period SHALL NOT start before the validity period and SHALL NOT end after the validity period of the corresponding Signer List Signer certificate.

9.9 Creation of Signer Lists

Any system that is creating Signer Lists SHALL use a private key corresponding to a Signer List Signer certificate that fulfills the criteria mentioned in ▶Section 9.8.

9.10 Processing of Signer Lists

Any system that is processing a Signer List with the intent to directly or indirectly use signer certificates from that Signer List for a document inspection SHALL check that list before processing it. This check SHALL use the corresponding Signer List Signer certificate. The system SHALL verify that the Signer List fulfills the following criteria:

- has a valid signature,
- the signature is from an eligible Signer List Signer (see ▶Section 9.8),
- the signing time is within the private key usage period of the Signer List Signer certificate

If any of the criteria is not fulfilled, the system SHALL NOT use that Signer List.

Appendix A Infrastructures for Terminals

A terminal is a reader component that contains the PCD used for communication between terminal and MRTD.

More than one reader component may be part of a single terminal. For example, all readers at the border control of an airport can be part of the same IS.

Even if the terminal consists of more than one reader, the terminal has only one identity. Regardless at which reader of a terminal a MRTD is presented, the identical key pair and CV certificates are used for TA between the terminal and the MRTD. From this point of view, the TA is done rather between the terminal and the MRTD than between the single reader and the MRTD.

A.1 Integrated Terminals

An overview over the architecture of an integrated terminal is given in ►Figure A.1.

An integrated terminal has only one reader. The HSM containing the private key of the terminal is physically part of the reader.

All communication between the HSM and the presented MRTD for performing the TA is done in the reader internally or between the PCD and the MRTD. No online connection to other components is needed for the TA.

The reader (or rather the terminal) needs an online connection to the DV to which it is associated. It must be connected only temporarily for requesting a new CV certificate or for downloading CV certificates of the DV or of CVCAs.

The disadvantage of this architecture is, that a stolen reader can be used to perform TA at least as long as the current CV certificate is valid. For this reason, organisational and/or technical security arrangements must be implemented within the intended operational environment of the terminal to avoid unauthorised access to the reader.

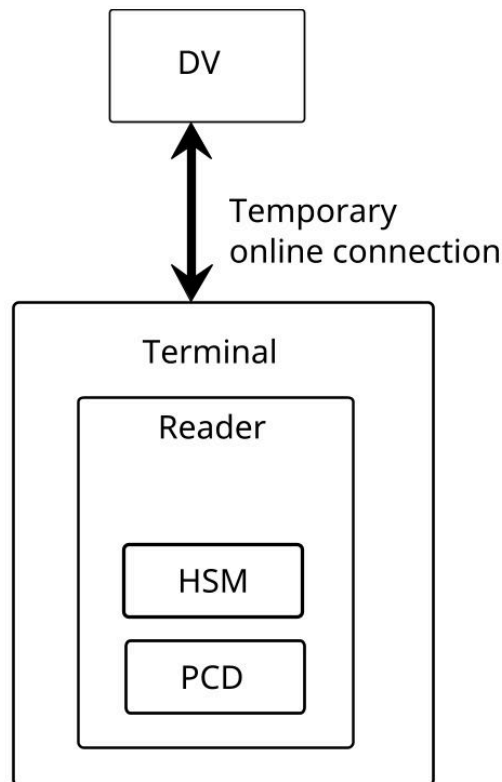


Figure A.1. Integrated terminal

A.2 Distributed Terminals

A distributed terminal may have more than one reader that is coordinated by a TCC. The HSM containing the private key of the terminal is not part of the readers. Instead the HSM is integrated into the TCC.

We distinguish between:

- distributed terminal with local control centre (see ▶Section A.2.1) and
- distributed terminal with remote control centre (see ▶Section A.2.2).

In a distributed terminal, all readers of a terminal have got a permanent online connection with the TCC. Since the private key of the terminal is stored in the HSM within the TCC, each TA between a terminal and a MRTD presented at one of its readers needs also some communication between the reader and the TCC.

The communication over the online connection between the TCC and the readers of the terminal must be secured. The actual measures to be taken to achieve the required level of security are decided by the operator of the terminal based on the operational environment. In any case it must be assured that the TCC does not communicate with a reader which is not under the control of the operator of the terminal.

A.2.1 Local Control Centre

For a distributed terminal, the connected readers do not need a (direct) online connection to the DV. Instead, the TCC performs all communication with the DV by requesting new CV certificates or downloading CV certificates of the DV or the CVCAs. For this, the online connection between the TCC and the DV is only temporarily required.

Hence, the advantage of this architecture is, that a stolen reader cannot be used for TA. Therefore, each reader can be operated easily in an insecure environment. Only for the TCC organisational and/or technical security arrangements must be implemented within the intended operational environment of the terminal to avoid unauthorised access to the TCC (especially to the HSM).

An overview over the architecture of a distributed terminal is given in ►Figure A.2.

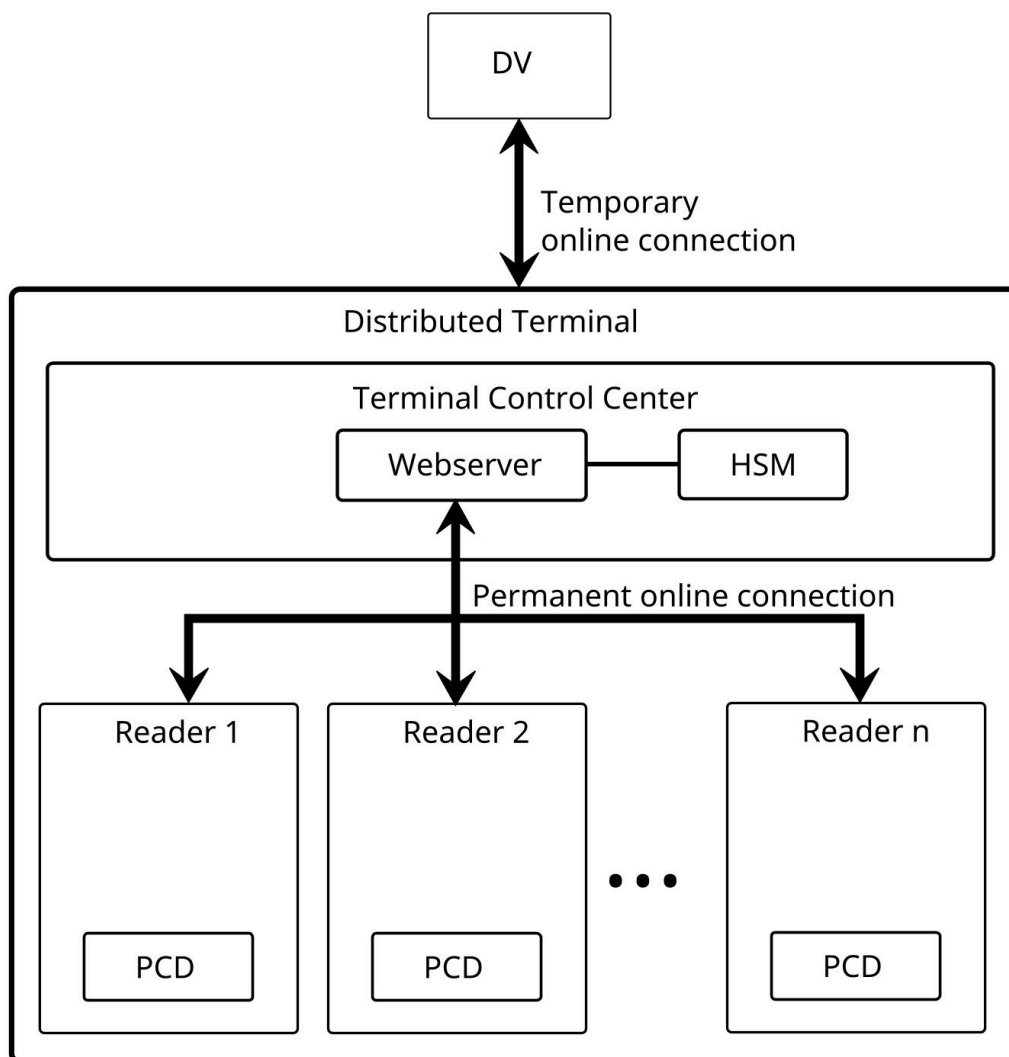


Figure A.2. Distributed terminal with local control centre

A.2.2 Remote Control Centre

If the TCC cannot be operated securely in the intended operational environment of the terminal, the TCC can also be operated in the operational environment of the DV provider.

In this case, a permanent online connection between the DV and each reader of the terminal is required for exchanging data during TA. However, the communication between the DV and the TCC for requesting a new CV certificate or for downloading CV certificates from the DV or of CVCAs can be implemented as DV-internal communication.

For this second variant of a distributed terminal, the same security aspects hold as mentioned above for the first variant.

An overview over the architecture of a distributed terminal of this second variant is given in ►Figure A.3.

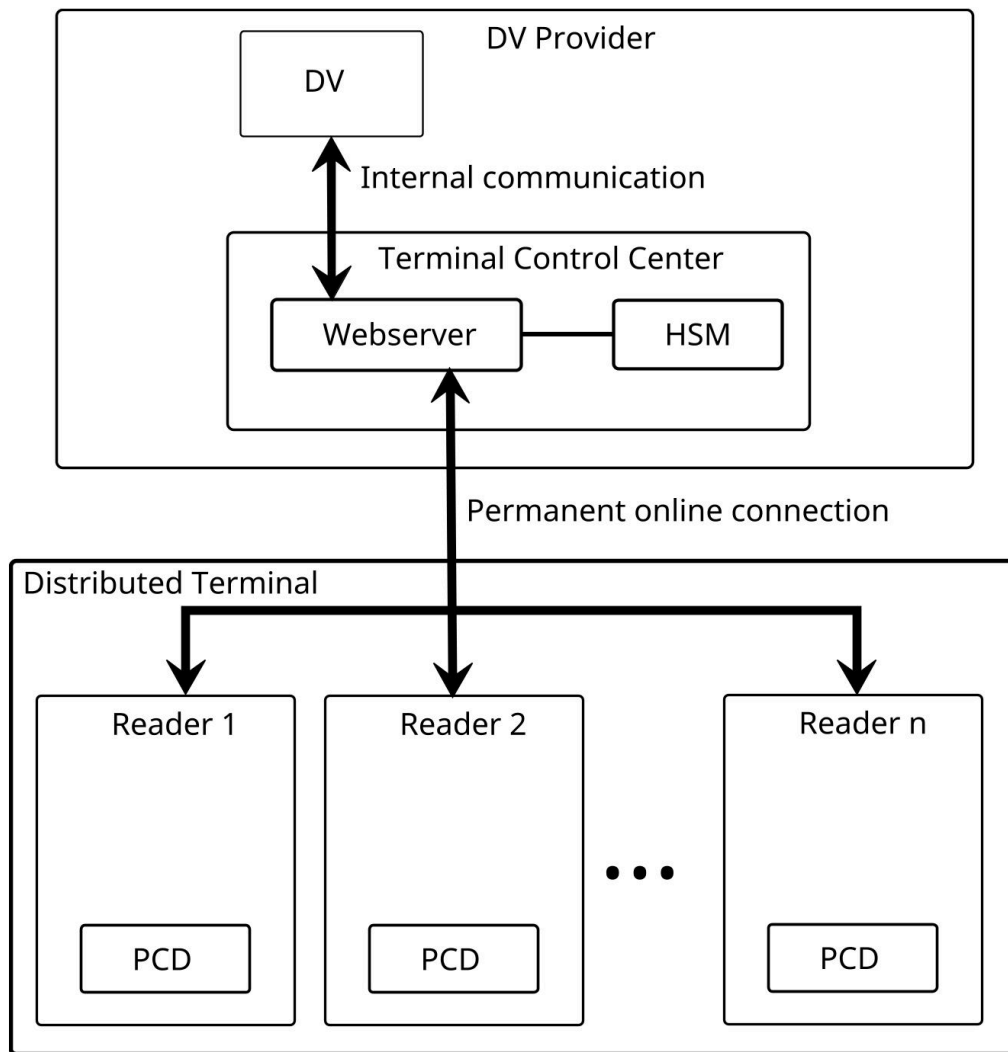


Figure A.3. Distributed terminal with control centre operated by the DV provider

List of Abbreviations

Abbreviation	Description
BER	Basic Encoding Rules
CA	Certificate Authority
CAR	Certificate Authority Reference
CHR	Certificate Holder Reference
CRL	Certificate Revocation List
CSCA	Country Signing Certificate Authority
CV	Card Verifiable
CVCA	Country Verifying Certificate Authority
CWT	CBOR Web Token
DFL	Defect List
DS	Document Signer
DTA	Digital Travel Authorization
DV	Document Verifier
EAC	Extended Access Control
EF.SOD	Elementary File Document Security Object
eID	Electronic Identity Document
eMRTD	Electronic Machine Readable Travel Document
HSM	Hardware Security Module
IC	Integrated Circuit
ICAO	International Civil Aviation Organization
IDB	ICAO Datastructure for Barcode
IS	Inspection System
LDS	Logical Data Structure
ML	Master List
MRTD	Machine Readable Travel Document
MRZ	Machine Readable Zone
NPKD	National Public Key Directory
OID	Object Identifier
PA	Passive Authentication
PCD	Proximity Coupling Device
PKI	Public Key Infrastructure
SKI	Subject Key Identifier
SOAP	Simple Object Access Protocol
SPOC	Single Point of Contact
TA	Terminal Authentication
TCC	Terminal Control Centre
TR	Technical Guideline
VDS	Visible Digital Seal
VDS-NC	Visible Digital Seal for Non-Constrained Environments

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